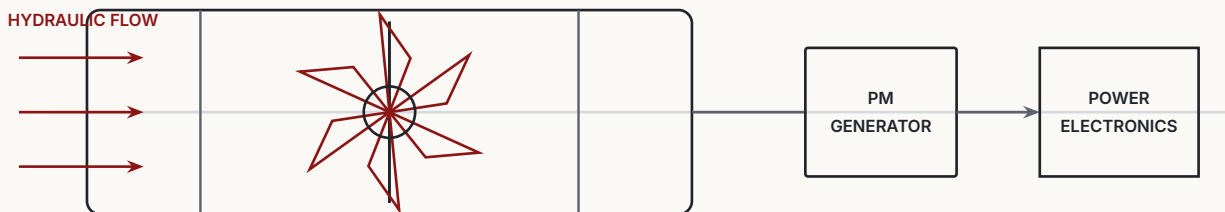


TURBOMACHINERY AND ENERGY CONVERSION

Integrated Design of a Compact Axial Hydraulic Turbine

Hydraulic sizing, CAD and structural verification, CFD,
electromechanical conversion, prototyping, and industrialisation



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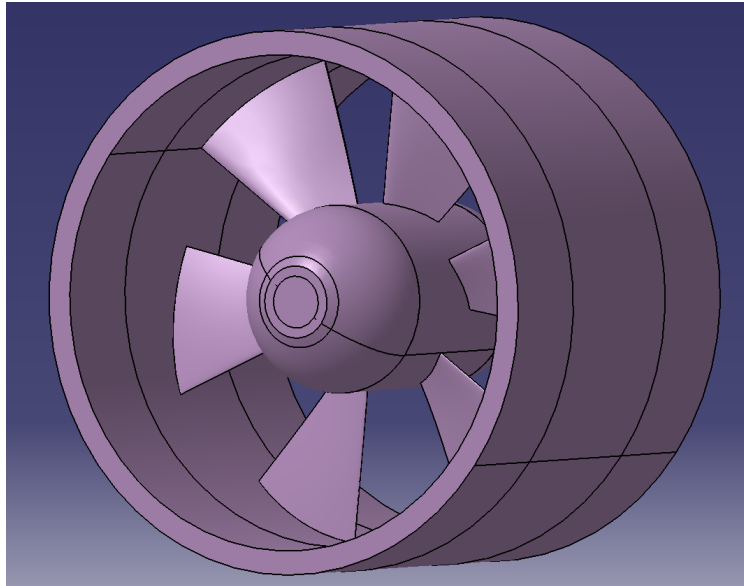
Abstract

This report presents an integrated engineering study of a compact axial hydraulic turbine, from similarity-based hydraulic sizing to prototype testing and industrial deployment. The runner is pre-sized in MATLAB from a target speed of 1500 rpm, a specific-speed class $N_{sq} = 200$, a design head of 3.8 m, and an outer radius of 0.103 m. Velocity triangles and a modified NACA thickness law define the blade sections, which are exported to CATIA V5 for surface reconstruction, full rotor assembly, and preliminary finite-element verification. The resulting fluid domain is analysed in STAR-CCM+ using a moving reference frame and a baseline mesh of 61,518 cells. The reported optimum produces approximately 1760 W at $0.144 \text{ m}^3 \text{ s}^{-1}$, below the initial 5 kW target and therefore motivating geometry refinement or a multi-unit architecture.

A surface-mounted permanent-magnet synchronous generator is then pre-sized from magnetic-circuit, induced-voltage, torque, and thermal relations. The associated voltage-source converter is examined through intersective PWM, carrier-frequency sensitivity, and closed-loop d - q current control. A PLA runner is manufactured by fused-filament fabrication and evaluated in a wind tunnel as a reduced-scale analogue. The experiment identifies a maximum power coefficient of approximately 0.239 and a peak efficiency of approximately 0.56. Finally, a modular industrial scenario based on 167 units is developed, including production planning, make-or-buy decisions, location strategy, and profitability. Under the source assumptions, the stated business case yields a 21.5% return on investment and a 2.4-year payback period. The study demonstrates the value of maintaining continuity between fluid mechanics, mechanical design, electrical conversion, experimental validation, and industrial decision-making.

KEYWORDS axial hydraulic turbine; Kaplan runner; Cordier diagram; STAR-CCM+; permanent-magnet synchronous generator; PWM converter; additive manufacturing; industrialisation.

Project at a Glance



Ducted axial-turbine assembly developed in CATIA V5 and used to prepare the computational fluid domain.

HYDRAULIC DESIGN

$N = 1500$ rpm, $H = 3.8$ m, $q_v = 0.132$ m³ s⁻¹, $R_e = 0.103$ m, and $R_i = 0.03914$ m.

DIGITAL ENGINEERING

MATLAB/XFoil blade synthesis, CATIA V5 surface and solid modelling, structural screening, and STAR-CCM+ moving-reference-frame CFD.

CFD BASELINE

61,518 cells, 279,251 faces, and 187,548 vertices; reported optimum $P \approx 1760$ W at $q_v \approx 0.144$ m³ s⁻¹.

EXPERIMENTAL RESULT

Additively manufactured PLA prototype; maximum measured power coefficient $C_p \approx 0.239$ and peak efficiency $\eta \approx 0.56$.

INDUSTRIAL SCENARIO

167 modular units, approximately 2 TWh/year target output, stated ROI 21.5%, and stated payback 2.4 years.

The equations, figures, numerical outputs, and engineering assumptions are retained from the original project report. The English prose, hierarchy, captions, scientific discussion, and publication design have been comprehensively edited for clarity and professional presentation.

Nomenclature

N, ω	Rotational speed and angular velocity.
N_{sq}, Ω	Engineering specific speed and dimensionless specific angular velocity.
H, q_v	Net head and volume flow rate.
R_e, R_i, r	Runner outer radius, hub radius, and local radius.
C_a, U, C_u	Axial velocity, blade peripheral speed, and whirl-velocity component.
$\beta_1, \beta_2, \alpha_2$	Blade inlet angle, blade outlet angle, and absolute discharge angle.
η, C_p	Efficiency and power coefficient.
P, M, C	Power and mechanical/electromagnetic torque.
B, H, μ	Magnetic flux density, magnetic field strength, and permeability.
p	Number of pole pairs.
e_a, e_b, e_c	Phase electromotive forces.
i_d, i_q	Direct- and quadrature-axis current components.
V_{dc}, f_c	DC-link voltage and PWM carrier frequency.
L, R, τ	Electrical inductance, resistance, and current-loop time constant.
$C_p = 2P/(\rho S V^3)$	Aerodynamic power coefficient used in the wind-tunnel analogy.

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Introduction

Hydraulic turbines have converted the energy of flowing water into useful mechanical power for centuries. From early water wheels to modern axial-flow machines, their development has been driven by the same engineering objective: extracting energy efficiently while maintaining structural integrity and operational reliability. In the present context of decarbonisation, distributed generation, and grid resilience, small and medium hydroelectric systems remain particularly attractive because they combine a mature conversion principle with low operational emissions and long service life.

This project addresses the end-to-end design of a compact axial hydraulic turbine and its associated electromechanical conversion chain. The work deliberately spans several engineering disciplines. Hydraulic preliminary sizing is first performed in MATLAB using similarity parameters and the Cordier diagram. Blade sections are then generated with XFOIL-derived profiles and transferred to CATIA V5 for three-dimensional modelling and structural verification. The complete flow domain is analysed in STAR-CCM+ to assess the velocity and pressure fields, mechanical power, efficiency, and cavitation risk. A permanent-magnet synchronous generator and its power-electronic conversion chain are subsequently pre-sized and studied. Finally, a polymer prototype is manufactured by fused-filament fabrication, tested in a wind tunnel as an aerodynamic analogue, and embedded in an industrial and commercial deployment study.

The study is therefore organised around six coupled work streams:

1. **Hydraulic preliminary design** - definition of the operating point, characteristic radii, velocity triangles, blade angles, and section thickness law.
2. **Blade and turbine CAD** - point-cloud export, surface generation, finite-element verification, and construction of the complete rotor and fluid domains.
3. **Computational fluid dynamics** - mesh generation, moving-reference-frame simulation, convergence assessment, performance extraction, and cavitation screening.
4. **Electromechanical conversion** - preliminary design of a surface-mounted permanent-magnet synchronous generator and analysis of the PWM-fed variable-speed drive.
5. **Prototype manufacture and experimental validation** - additive manufacture, wind-tunnel measurements, and identification of the power coefficient and efficiency.
6. **Industrialisation** - market positioning, production planning, resource requirements, make-or-buy decisions, location strategy, and financial viability.

The resulting document is intended as a coherent engineering publication rather than a sequence of isolated laboratory exercises. Assumptions are stated explicitly, equations are retained in their original mathematical form, and each numerical stage is connected to the physical design decisions that follow.

Hydraulic Preliminary Design

2.1 Design brief and operating point

The first stage establishes a hydraulic pre-design for an axial-flow turbine. The initial specification is:

- rotational speed: $N = 1500$ rpm;
- specific-speed class: $N_{sq} = 200$;
- maximum available net head: $H = 4$ m;
- penstock and runner outer radius: $R_e = 0.103$ m.

The process comprises three linked tasks: MATLAB-based sizing of the runner, construction of the blade geometry, and subsequent preparation of the complete fluid domain for CFD analysis. The MATLAB implementation is reproduced in the appendices.

2.2 Similarity parameters and Cordier selection

The angular velocity is

$$\omega = \frac{2 \times \pi \times N}{60} = 157.07 \text{ rad.s}^{-1}.$$

The dimensionless specific angular velocity is written as

$$\Omega = \frac{\omega \sqrt{q_v}}{(gH)^{3/4}} = \frac{\frac{\pi N}{30} \times \sqrt{q_v}}{(gH)^{3/4}} = \frac{\Phi^{1/2}}{\Psi^{3/4}},$$

$$Nsq = \frac{N \sqrt{q_v}}{H^{3/4}},$$

which gives

$$\Omega = \frac{\pi \cdot Nsq}{30 \cdot g^{3/4}} = 3.778.$$

The Kaplan region of the Cordier diagram was digitised with Engauge and imported into MATLAB. This provided a reproducible estimate of the specific radius rather than a purely graphical reading.

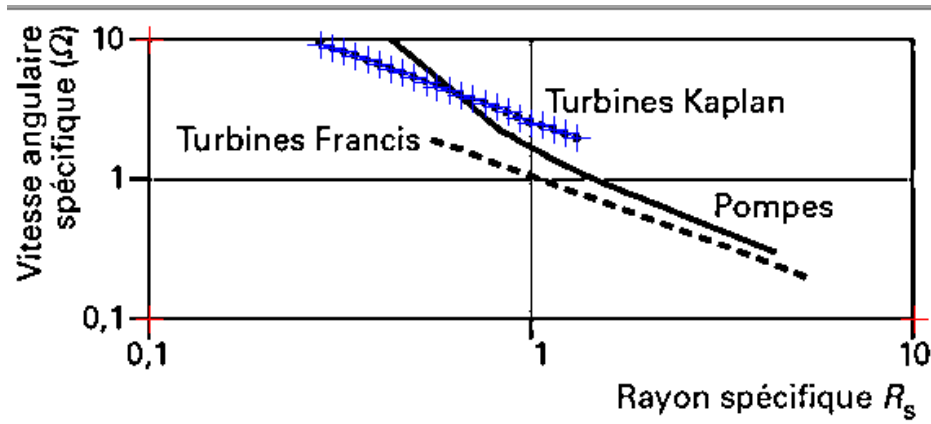


Figure 2.1: Digitised Kaplan-turbine region of the Cordier diagram.

The linear fit used in the selected interval is

$$\Omega(R_s) = \frac{9.224 - 1.987}{0.2804 - 1.31} R_s + 8.705 = -7.03 R_s + 8.70.$$

Solving $\Omega(R_s) = 3.778$ gives

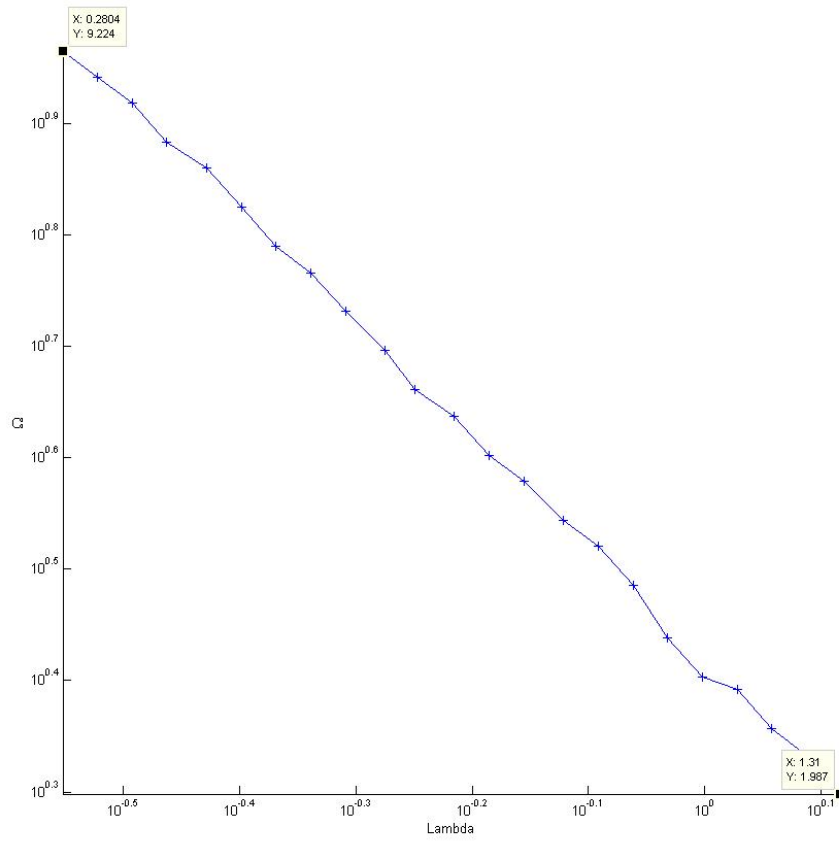


Figure 2.2: MATLAB interpolation of the Cordier data used to identify the specific radius.

$$R_s = 0.701.$$

2.3 Head, flow rate, and runner radii

The head and volume-flow relations are

$$\Omega = \frac{\omega \sqrt{q_v}}{(gH)^{\frac{3}{4}}}, \quad \Lambda = \frac{R(gH)^{\frac{1}{4}}}{\sqrt{q_v}}.$$

The selected operating values are

$$H = 3.8 \text{ m}$$

and

$$Nsq = \frac{N \sqrt{q_v}}{H^{\frac{3}{4}}},$$

$$q_v = 0.132 \text{ m}^3 \cdot \text{s}^{-1}.$$

The hub radius is defined from the hub-to-tip ratio $T = 0.38$:

$R_i = R_e T$, with $R_e = 0.1030 \text{ m}$, hence

$$R_i = 0.03914 \text{ m}.$$

The axial velocity component is

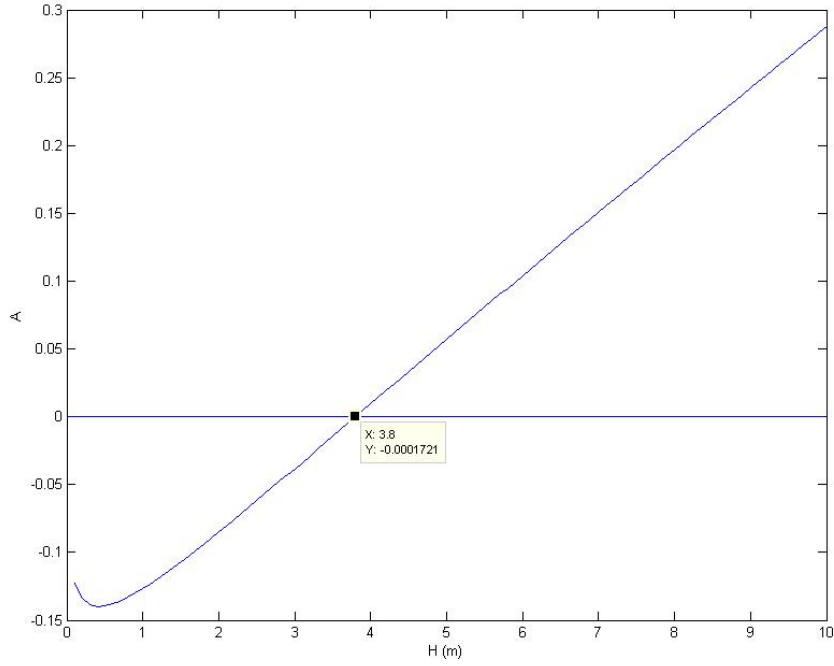


Figure 2.3: Graphical solution used to determine the design head.

$$C_a = \frac{q_v}{S} = \frac{q_v}{\pi(R_e^2 - R_i^2)},$$

$$C_a = 4.629 \text{ m.s}^{-1}.$$

The circumferential speed and theoretical and corrected whirl components are evaluated over $R_i < r < R_e$.

Rayon (r) [m]	U(r) [m/s]	Cu_th(r) [m/s]	Cu(r) [m/s]
0,039	6,148	5,904	6,418
0,046	7,262	4,998	5,433
0,053	8,377	4,333	4,710
0,060	9,491	3,824	4,157
0,068	10,606	3,422	3,720
0,075	11,720	3,097	3,366
0,082	12,835	2,828	3,074
0,089	13,949	2,602	2,828
0,096	15,064	2,410	2,619
0,103	16,178	2,244	2,439

Figure 2.4: Radial distribution of blade speed and whirl-velocity components.

2.4 Velocity triangles and blade mean line

The inlet, outlet, and discharge angles ($\beta_1, \beta_2, \alpha_2$) are calculated directly in MATLAB from the local velocity triangles.

The calculated values of β_1 and β_2 remain between 30° and 80° , which is consistent with the adopted axial-runner design range. The blade-section mean line is generated by the function `LigneK.m`. Assuming a linear evolution of the local tangent between the inlet and outlet angles,

$$d_{yy} = (\tan \beta_2 - \tan \beta_1) x_{x2} + \tan \beta_1.$$

Command Window			
Re (m), Ri (m)			
0.1030	0.0390		
4.6125			
verif.:r(u)	beta_1	beta_2	alph_2
0.0390	53.0231	68.5249	-39.4854
0.0550	61.9027	69.9072	-49.2833
0.0710	67.5311	72.0379	-56.3088
0.0870	71.3496	74.0844	-61.4501

Figure 2.5: MATLAB output for inlet, outlet, and discharge angles at selected radii.

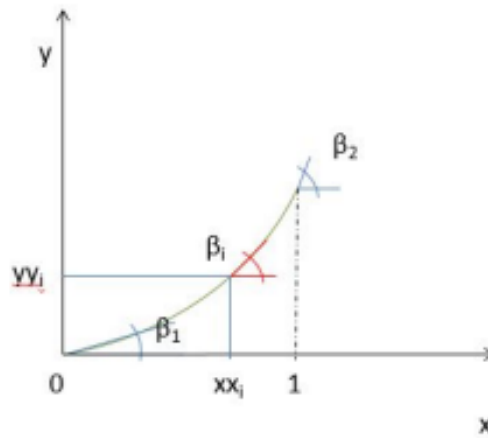


Figure 2.6: Construction of the blade-section mean line from the local inlet and outlet angles.

The full MATLAB routine is provided in Appendix A. It returns both the mean-line ordinate and the local blade angle required for the subsequent thickness distribution.

2.5 Thickness law and modified NACA section

Mechanical strength requires a blade that is thick near the hub and progressively thinner toward the tip. The adopted relative-thickness law is therefore defined as a function of radius.

Standard NACA profiles were found to be too thin for the intended prototype and structural constraints. A thicker profile was generated and fitted to a modified polynomial thickness law using XFOIL data and MATLAB optimisation.

In the reconstruction, the red curve represents the upper surface, the green curve the normalised thickness law, and the blue curve the original point set. The function `fminsearch(f, x0)` is used to minimise the quadratic discrepancy between the measured and analytical profiles.

The resulting section is scaled according to the local solidity, defined here as the chordwise spacing relationship between adjacent blades. The final runner geometry is shown below.

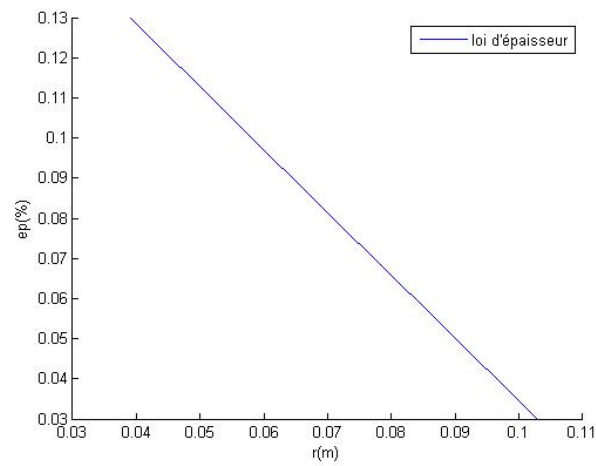


Figure 2.7: Relative blade-thickness law from hub to tip.

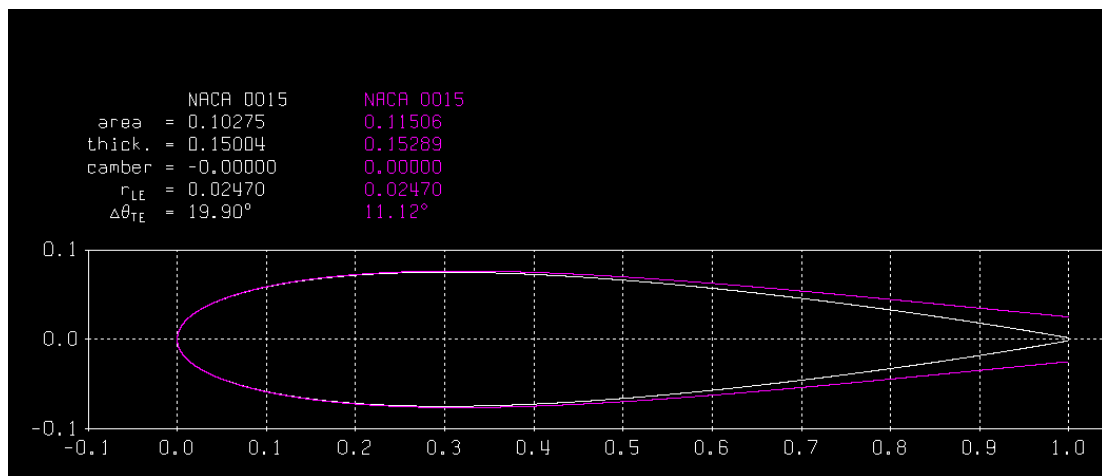


Figure 2.8: Comparison between the initial NACA profile and the modified, thicker section.

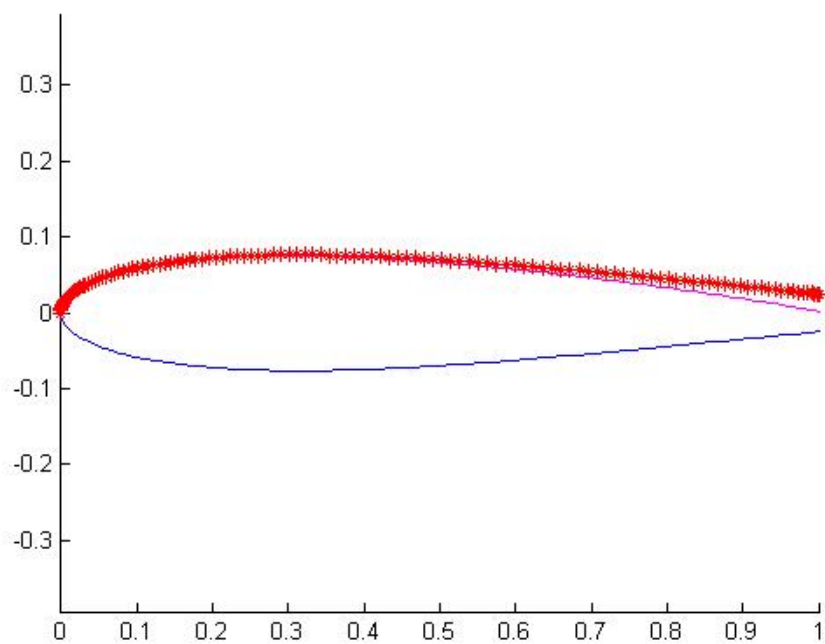


Figure 2.9: MATLAB reconstruction of the blade section from raw points and the fitted thickness law.

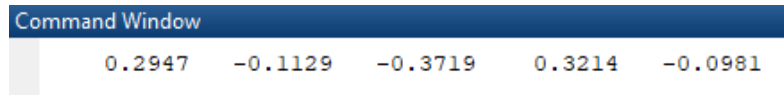


Figure 2.10: Optimised polynomial coefficients returned by MATLAB.

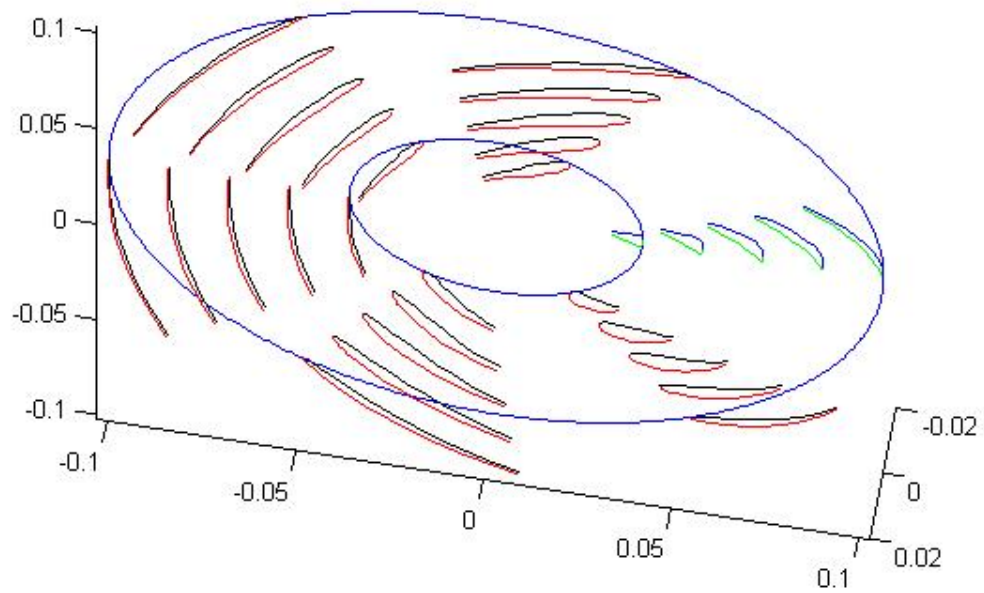


Figure 2.11: Final blade-row geometry obtained after scaling and circumferential replication.

CAD Modelling and Structural Verification

3.1 Point export and surface generation

Once the analytical blade geometry had been defined, MATLAB was used to decompose each section into a set of three-dimensional points. These point clouds were exported to CATIA V5, where the blade surfaces were reconstructed with the Generative Shape Design workbench. This workflow preserves the radial evolution of chord, thickness, and stagger angle while providing a CAD model suitable for both structural analysis and additive manufacture.

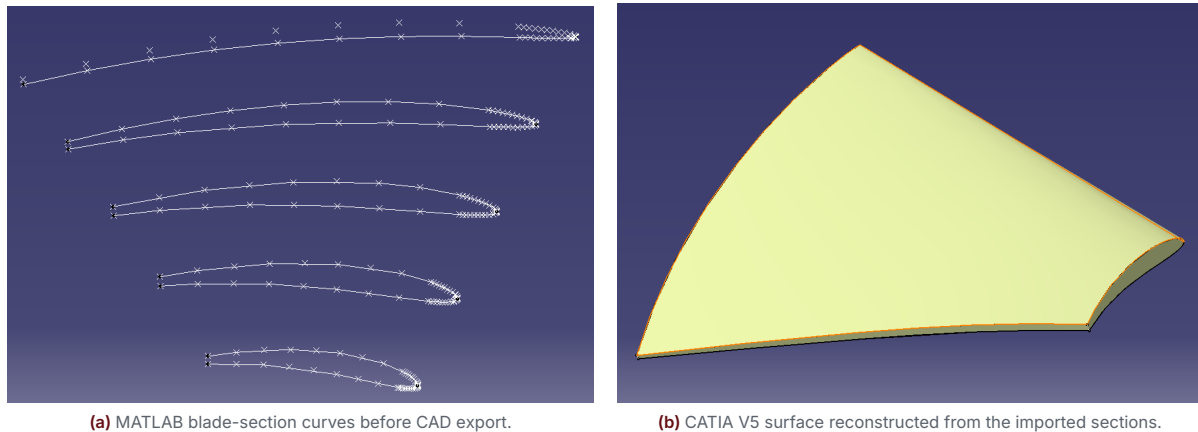


Figure 3.1: Transfer of the analytical blade definition into the CAD environment.

3.2 Finite-element verification of the blade

A preliminary finite-element analysis was performed to verify that the modelled blade could withstand the distributed fluid loading. Surface tractions were applied to the pressure and suction sides, while the blade root was fixed. The highest equivalent stresses occur, as expected, near the root constraint.

The maximum calculated stress remains below the elastic limit of the selected material within the assumptions of the model. The analysis therefore supports continuation to the full rotor assembly, while also identifying the root transition as the principal area for future mesh and geometry refinement.

3.3 Rotor, hub, and duct assembly

The individual blade was replicated circumferentially using the CATIA Part Design circular-pattern tool.

The hub was then modelled so that its outer diameter remained compatible with the blade-root envelope.

A surrounding duct was divided into three axial regions. This partitioning was introduced specifically to define the inlet, rotating, and outlet fluid-domain interfaces required by STAR-CCM+.

A two-dimensional drawing was generated to retain the principal dimensions and assembly envelope.

Finally, the solid geometry was used to construct the fluid volumes for CFD.

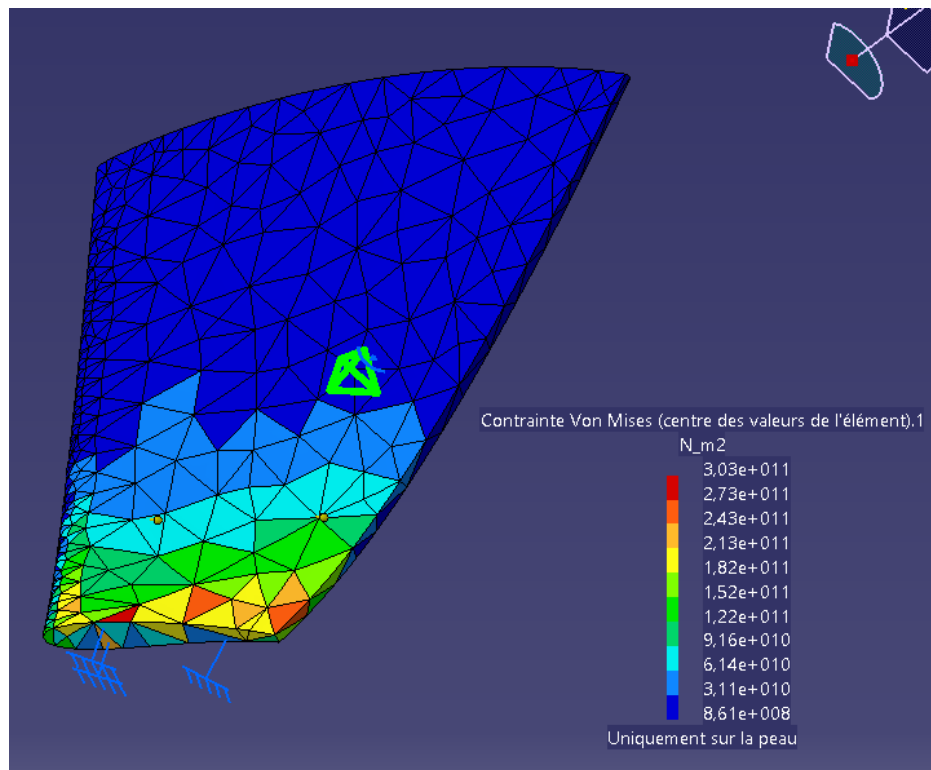
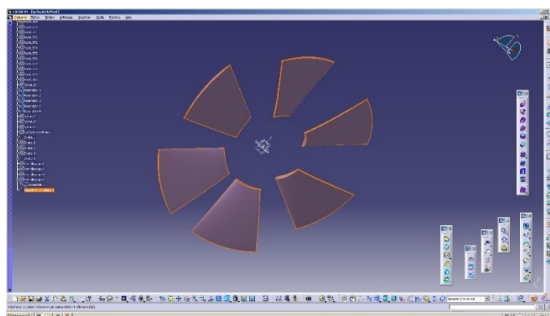
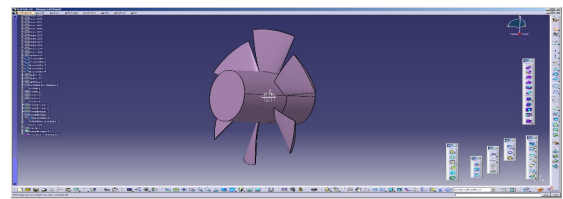


Figure 3.2: Equivalent-stress distribution in the blade under the prescribed surface loading.



(a) Circular blade replication around the runner axis.



(b) Hub and blade integration in CATIA V5.

Figure 3.3: Progressive construction of the complete runner.

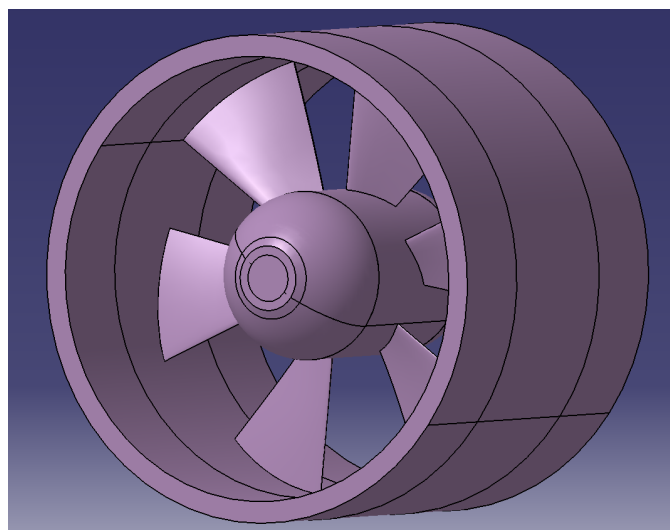


Figure 3.4: Complete ducted-turbine CAD assembly.

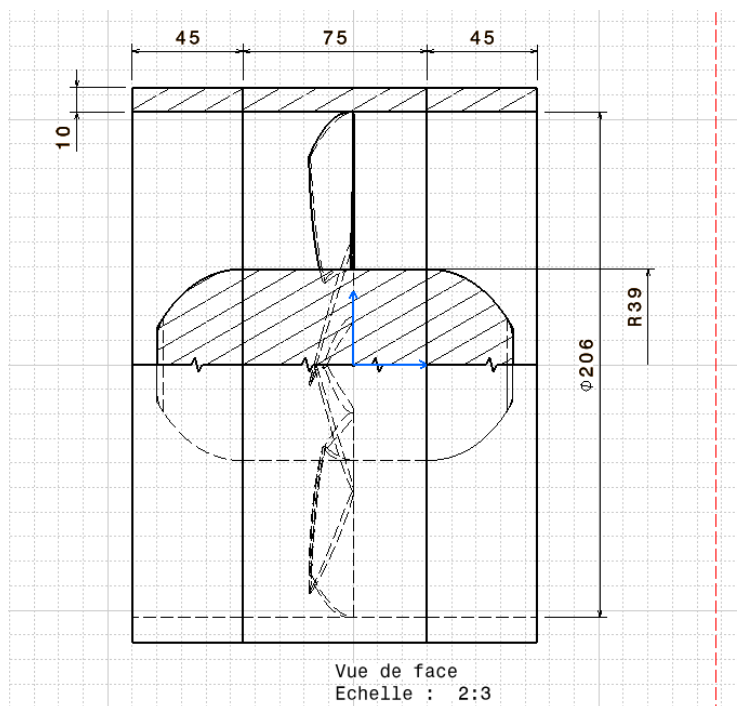


Figure 3.5: Principal CAD dimensions of the ducted runner.

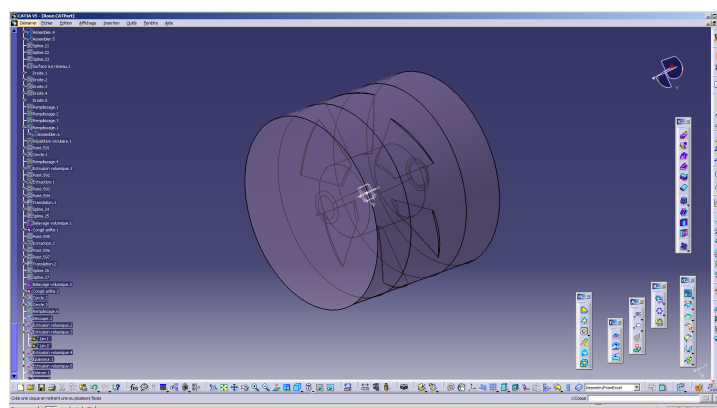


Figure 3.6: Fluid-domain volumes extracted from the complete CAD assembly.

Computational Fluid-Dynamics Analysis

4.1 Numerical model and mesh

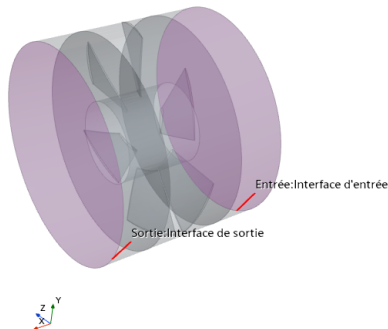
The STAR-CCM+ study was configured to verify the following design targets:

- $N = 1500$ rpm;
- $N_{sq} = 200$;
- $H = 4$ m;
- $R = 0.103$ m;
- target hydraulic efficiency $\eta = 0.92$;
- target power $P = 5$ kW.

The inlet and outlet surfaces were defined at the ends of the extended flow domain. Additional upstream and downstream lengths were retained to prevent the solution from being dominated by the immediate blade-wake structures.

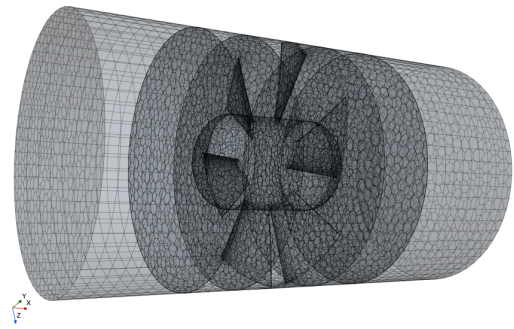
A combined surface and volume mesh was generated.

Simcenter STAR-CCM+



(a) Inlet, rotating-domain interfaces, and outlet.

Simcenter STAR-CCM+



(b) Surface and volume mesh of the fluid domain.

Figure 4.1: Computational-domain definition and baseline discretisation.

The baseline mesh contains 61,518 cells, 279,251 faces, and 187,548 vertices.

4.2 Physical models and reference frame

A segregated-flow formulation was selected to reduce computational cost and memory demand while maintaining acceptable convergence for the available computing resources. Rotor motion was represented with a moving reference frame. This formulation permits the rotating region to be treated in a steady framework while preserving the apparent inertial effects associated with rotation. The reference pressure was set to 101,300 Pa.

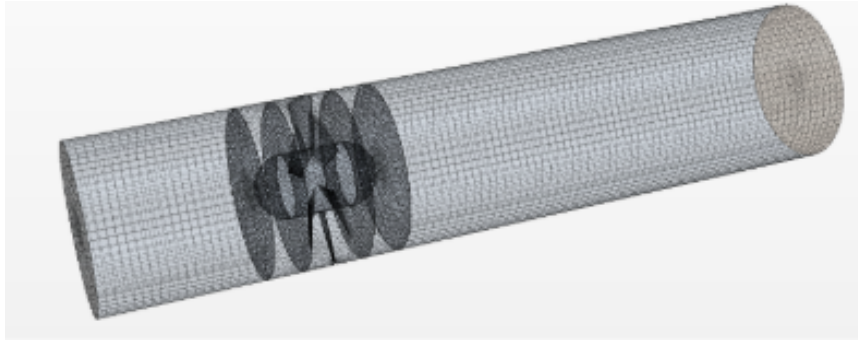


Figure 4.2: Detail of the computational mesh in the runner region.

4.3 Convergence assessment

The stopping criterion was set to 1000 iterations, with the residual level expected to fall below 10^{-3} . The residual history confirms that this iteration count provides a sufficiently settled baseline solution for the present comparative design study.

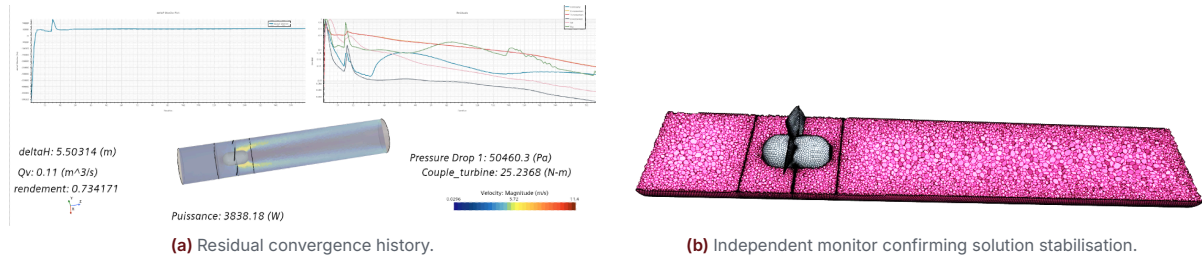


Figure 4.3: Convergence assessment of the baseline CFD computation.

If convergence were not achieved, the appropriate response would be to revisit the boundary conditions, pressure level, mesh quality, turbulence treatment, or iteration budget rather than interpreting an unconverged field.

4.4 Performance extraction and interpretation

The flow-field results were post-processed to obtain the hydraulic torque, mechanical power, and efficiency trends.

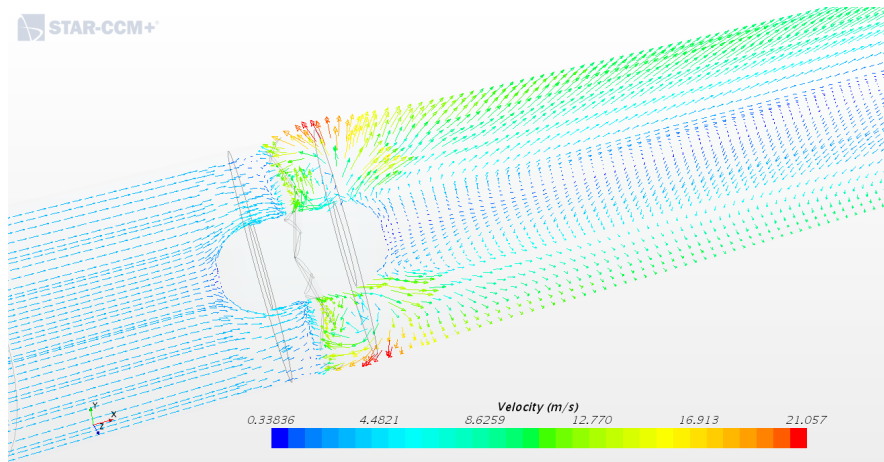


Figure 4.4: Representative CFD field used for performance interpretation.

The available mechanical power is calculated from

$$P_m = C\Omega = M\Omega.$$

N (RPM)	Qm (kg/s)	Rendement (%)	DeltaP (Pa)	DeltaP (bar)	DeltaH (m)	Moment (N.m)	Omega	Rho	Pm (W)	Qv (m³/s)
1500	111,1	67,67	12019	0,12019	1,2221	82,82	158,57	1010	1300,274	0,1111
1500	128,27	68,68	23836	0,23836	2,3836	99,586	158,57	1010	1563,48	0,12827
1500	131,3	71,71	26664	0,26664	2,6664	102,01	158,57	1010	1601,557	0,1313
1500	141,4	74,74	36057	0,36057	3,6057	112,11	158,57	1010	1760,127	0,14443
1500	146,45	72,72	41511	0,41511	4,1511	117,16	158,57	1010	1839,412	0,14645
1500	151,5	69,69	47470	0,4747	4,747	122,21	158,57	1010	1918,697	0,1515
1500	156,55	68,68	52520	0,5252	5,252	127,26	158,57	1010	1997,982	0,15655
1500	161,6	65,65	58479	0,58479	5,8479	132,31	158,57	1010	2077,267	0,1616

Figure 4.5: Performance curves extracted from the STAR-CCM+ operating-point sweep.

The best simulated operating point occurs at approximately $q_v = 0.144 \text{ m}^3 \cdot \text{s}^{-1}$, with a mechanical power of about 1760 W and a corresponding head of $H = 3.60 \text{ m}$. This result does not reach the initial 5 kW target for a single unit, but it indicates that prioritising efficiency and using two units may be preferable to forcing one runner toward a less favourable operating point.

Cavitation becomes possible when the local static pressure falls below the saturation pressure. The highest-risk regions are therefore expected near leakage paths and blade edges, where local velocity peaks produce the strongest pressure depression.

The numerical study identifies four principal improvement directions:

- refine the blade-row geometry to optimise the balance between solidity, loading, and friction;
- increase mesh resolution in the blade boundary layers, tip region, and wake;
- revisit the single-unit 5 kW objective and evaluate a multi-unit architecture;
- adopt a less optimistic efficiency assumption during early-stage sizing.

Within the limits of the mesh and steady moving-reference-frame model, the CFD study completes the hydraulic pre-design and provides the loading and operating-point information required for the electrical and prototype stages.

Electromechanical Conversion System

5.1 Preliminary design of a permanent-magnet synchronous generator

The hydraulic runner is coupled to a surface-mounted permanent-magnet synchronous machine operated as a generator. The purpose of this section is to establish a first set of electromagnetic dimensions at the selected operating point and to derive the relationships governing induced voltage, electromagnetic torque, magnet loading, and active length.

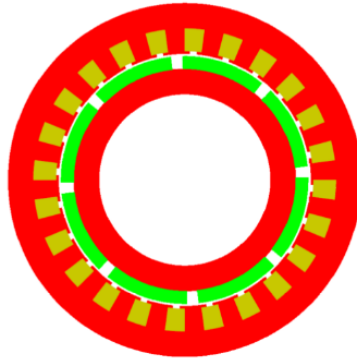


Figure 2 : Exemple de machine à pôles lisses ($p=4$)

Figure 5.1: Cross-section of the permanent-magnet synchronous machine used for the preliminary model.

The magnets are mounted on the rotor surface, surrounding the previously designed turbine. The ferro-magnetic laminations form high-permeability magnetic circuits, while the phase windings are placed in stator slots.

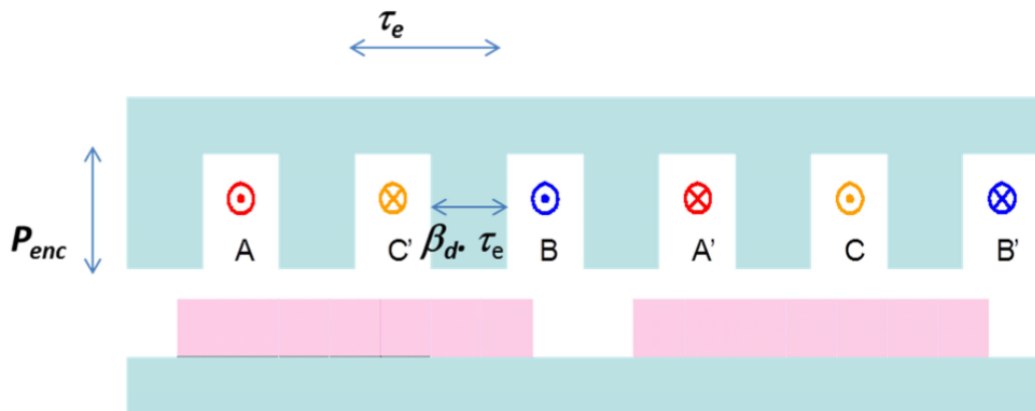


Figure 5 : vue d'une paire de pôles de la machine

Figure 5.2: Winding layout, showing the outgoing and return conductors for each phase.

The end effects are neglected, allowing the machine to be analysed in two dimensions. The model therefore assumes an axially uniform machine in which the displayed section is representative of any section along the

active length. The permeability of the ferromagnetic yokes is treated as very high relative to the air gap and magnet regions.

5.1.1 Preliminary magnetic relations

For a linear medium,

$$\vec{B} = \mu \vec{H}.$$

For air,

$$\vec{B}_0 = \mu_0 \vec{H}_0,$$

and for the iron,

$$\vec{B} = \mu_0 \mu_r \vec{H}, \text{ with } \mu_r > 1000.$$

For the permanent magnet,

$$\vec{B} = \mu_0 \vec{H} + \vec{B}_r.$$

5.1.2 Electromotive forces and average torque

When the rotor turns relative to the stator, the magnet field induces phase electromotive forces. In a stator-fixed reference frame,

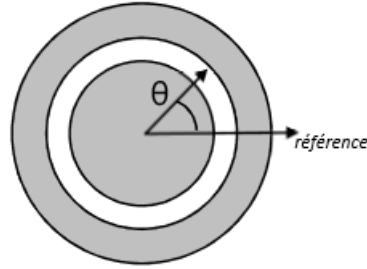


Figure 5.3: Stator-fixed angular reference used for the harmonic field representation.

$$B(\theta, t) = \sum_{k=0}^{\infty} B_{2k+1} \cos[p(2k+1)(\theta - \Omega t)],$$

where p is the number of rotor pole pairs and Ω is the mechanical angular velocity in radians per second. The flux-cutting relation is illustrated below.

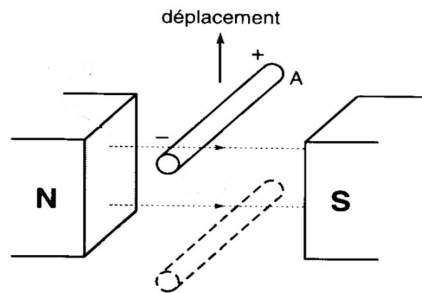


Figure 5.4: Flux-cutting principle for a conductor moving through a magnetic field.

For a conductor moving at velocity $\vec{v}$,

$$e = l |\vec{B} \wedge \vec{v}| = l B v \text{ if } \vec{B} \perp \vec{v}.$$

With $LxB = \phi$,

$$\frac{d\phi}{dt} = \frac{dx}{dt} LB = vLB.$$

For phase A, with D denoting the air-gap diameter,

$$e_{\text{condA}} = B(0, t) * \frac{\Omega D}{2} L,$$

$$e_{\text{condA}'} = B\left(\frac{\pi}{2}, t\right) \frac{\Omega D}{2} L = -e_{\text{condA}}.$$

Because one turn contains two active conductors, N conductors correspond to $N/2$ turns. Accounting for the pole count, the phase-A electromotive force becomes

$$e_{\text{phaseA}} = \Sigma e_{\text{condA}} - \Sigma e_{\text{condA}'} = nB(0, t) \frac{\Omega D}{2} L.$$

The two other phase voltages are shifted by $2\pi/3$ and $4\pi/3$:

$$e_{\text{phaseB}} = nB\left(\frac{2\pi}{3}, t\right) \frac{\Omega D}{2} L,$$

$$e_{\text{phaseC}} = nB\left(\frac{4\pi}{3}, t\right) \frac{\Omega D}{2} L.$$

Assuming balanced sinusoidal currents,

$$i_i(t) = I\sqrt{2} \cos\left(\omega t - \Psi - (i-1)\frac{2\pi}{3}\right), \quad \omega = p\Omega.$$

The instantaneous electromagnetic power is

$$P_{em}(t) = \sum_{i=1}^3 e_i i_i = n \frac{\Omega D}{2} L \sum_{i=1}^3 \sum_{k=0}^{\infty} B_{k+1} \cos\left[(2k+1)\left((i-1)\frac{2\pi}{3} - \omega t\right)\right] \cos\left(\omega t - \Psi - (i-1)\frac{2\pi}{3}\right).$$

All higher-order terms have zero mean over an electrical period; only the fundamental contributes to the average power:

$$\langle p_{em} \rangle = I\sqrt{2}n \frac{\Omega D}{2} L \sum_{i=1}^3 B_1 \cos\left((i-1)\frac{2\pi}{3} - \omega t\right) \cos\left(\omega t - \Psi - (i-1)\frac{2\pi}{3}\right),$$

$$\langle p_{em} \rangle = \frac{3}{2} I\sqrt{2}n \frac{\Omega D}{2} L \cos(\Psi).$$

Using the linear current loading $A = \frac{3nI}{\pi D}$ and bore volume $V = \frac{\pi D^2 L}{4} = \pi R^2 L$,

$$\langle C_{em} \rangle = \frac{\langle p_{em} \rangle}{\Omega} = \frac{3}{2\sqrt{2}} n B_1 I D L \cos \Psi = \frac{3}{2\sqrt{2}} \frac{A \pi D}{3} D B_1 L \cos \Psi,$$

$$\langle C_{em} \rangle = \sqrt{2} B_1 A V \cos(\Psi).$$

The maximum torque is obtained for $\Psi = 0$:

$$\langle C_{em} \rangle_{\max} = \sqrt{2} B_1 A V.$$

5.2 Magnet thickness and fundamental air-gap flux density

The air gap is assumed sufficiently small for the induction in the gap and the magnet to be predominantly radial. Curvature and load-current effects are neglected, and the ferromagnetic yokes are assumed to have infinite permeability. The magnet characteristic is

$$B = \mu_0 H + B_r.$$

Ampere's law is applied to a contour crossing the two magnets and the two air gaps:

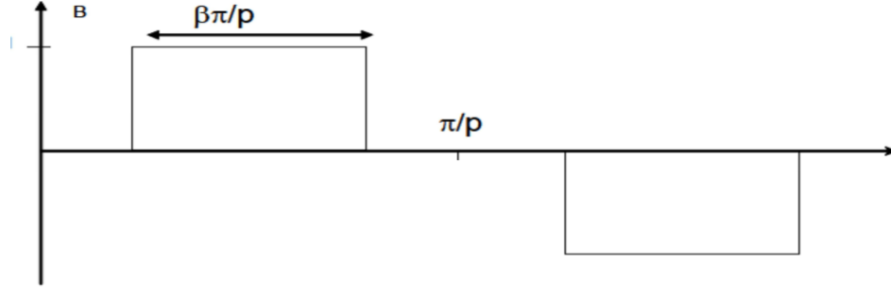


Figure 4 : forme d'onde de l'induction dans l'entrefer en fonction de θ dans le repère lié au rotor

Figure 5.5: Idealised radial air-gap flux-density waveform created by the surface magnets.

$$\text{rot}(\vec{H}) = \vec{J} \rightarrow \begin{cases} \vec{B} = \mu_0 \vec{H} \text{ in air} \\ \vec{B} = \mu_0 \mu_r \vec{H} \text{ in iron, } \mu_r > 1000 \\ \vec{B} = \mu_0 \vec{H} + \vec{B}_r \text{ in the magnet} \end{cases}$$

Because no current is enclosed,

$$\oint \vec{H} \cdot d\vec{l} = 0.$$

The field intensity in the iron is negligible compared with that in the air gap:

$$H_{fer} = \frac{B_{fer}}{\mu_0 \mu_r} \ll H_{air}.$$

The magnetic-circuit equation therefore reduces to

$$2H_a e_a + 2H_g g = 0, \quad B_a = B_g = \hat{B}.$$

With

$$\begin{cases} B_g = \mu_0 H_g \rightarrow H_g = \frac{\hat{B}}{\mu_0} \\ B_a = \mu_0 H_a + B_r \\ H_a = \frac{\hat{B} - B_r}{\mu_0} \end{cases},$$

one obtains

$$\hat{B} = \frac{e_a B_r}{g + e_a}.$$

For $g = 1 \text{ mm}$ and e_a varied from 0.1 to 10 mm, the relationship is initially nearly linear and asymptotically approaches B_r . Excessive magnet thickness therefore adds material with progressively smaller electromagnetic benefit.

The first harmonic of the rectangular air-gap field is

$$a_1 = \frac{2}{T} \int_{-T/2}^{T/2} f(T) \cos(\omega t) dt,$$

which gives

$$B_1 = a_1 = \frac{2}{2\pi} \int_{-\frac{\pi}{p}}^{\frac{\pi}{p}} B(\theta) \cos(p\theta) d\theta = \frac{4}{\pi} \hat{B} \sin\left(\frac{\beta\pi}{2}\right),$$

$$B_1 = \frac{4}{\pi} \hat{B} \sin\left(\frac{\beta\pi}{2}\right).$$

5.3 Dimensional model at the nominal operating point

The machine is sized for a nominal power and speed (P_n, Ω_n). The principal prescribed quantities are summarised below.

Symbol	Definition
J	RMS current density in the conductors (A/mm ²)
A_L	linear current loading
k_r	slot fill factor
D	air-gap diameter
$B_{e,max}$	maximum air-gap induction
B_r	magnet remanent flux density
g	air-gap length
B_{sat}	saturation flux density
f_{max}	maximum electrical frequency

The slot fill factor is

$$k_r = \frac{\sum S_{cond}}{S_{slots}},$$

where S_{cond} is the conductor cross-sectional area.

5.3.1 Thermal and electromagnetic constraints

Copper losses and convection establish the principal thermal constraint:

$$P_J = 3nR_{cond}I^2 = 3n\rho\frac{L}{S_{cond}}I^2 = \rho 3nILJ = \rho\pi DL AJ,$$

$$\Delta T \sim \frac{P_J}{hS_{ext}}, \quad S_{ext} \sim \pi DL,$$

$$\Delta T \sim \frac{\rho AJ}{h}.$$

Representative design ranges are

$$J \in [3 \times 10^6, 7 \times 10^6] \text{ A/m}^2,$$

$$A \in [12,000, 60,000] \text{ A/m}.$$

At the nominal point,

$$P_N = C_{em}\Omega_N, \quad C_{em} = \frac{P_N}{\Omega_N} = \frac{P_N}{\frac{\pi N}{30p}},$$

which gives

$$C_{em} = 152 \text{ Nm}.$$

The active length is selected while respecting magnetic saturation.

With

$$X = \frac{\hat{B}}{B_r},$$

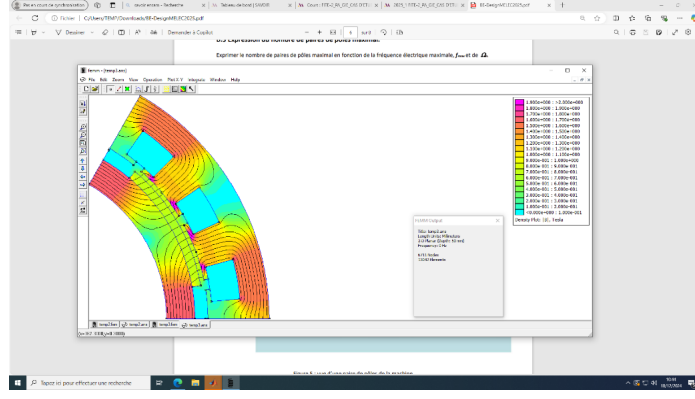


Figure 5.6: Sizing interface used to define the generator geometry and magnetic loading.

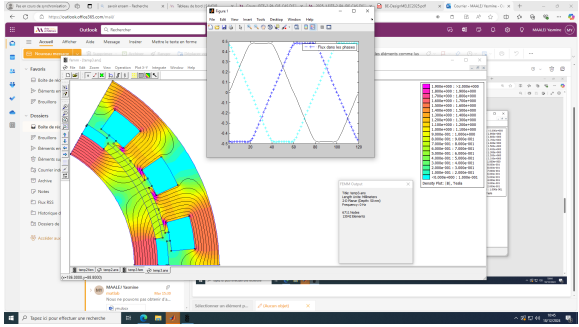
$$\hat{B} = XB_r, \quad B_1 = \frac{4}{\pi} \hat{B} \sin\left(\frac{\beta\pi}{2}\right),$$

$$C_{em} = \sqrt{2}AD^2LB_r \sin\left(\frac{\beta\pi}{2}\right),$$

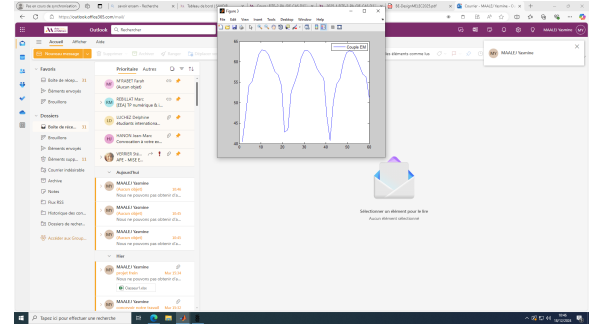
one obtains

$$L = \frac{C}{\sqrt{2}AXB_rD^2 \sin\left(\frac{\beta\pi}{2}\right)}.$$

The resulting parameter set is then introduced into the numerical model to calculate flux linkage as a function of rotor position and phase and to verify the electromagnetic design.



(a) Generator pre-sizing parameters and principal numerical results.



(b) Complementary machine-design output used for finite-element validation.

Figure 5.7: Numerical outputs of the preliminary generator-sizing model.

This stage provides a consistent first-order geometry for the permanent-magnet synchronous generator at the selected torque-speed point.

5.4 Variable-speed drive and PWM converter

The second electrical study concerns the conversion chain formed by the permanent-magnet synchronous machine and a voltage-source inverter controlled by pulse-width modulation (PWM). The objective is to understand the variable-speed control principle and evaluate the influence of switching and filtering parameters on the phase currents.

5.4.1 Global speed-control architecture

The rotating-system dynamics may be expressed in the generic form

$$J \frac{d\Omega}{dt} = C_{em} - C_{load},$$

where J is the total rotating inertia, Ω the rotational speed, C_{em} the electromagnetic torque, and C_{load} the

torque imposed by the turbomachine.

The control structure compares the speed reference with the measured speed, uses a PI/PID controller to produce a torque reference, and applies vector control so that the phase currents remain synchronised with the machine electromotive forces.

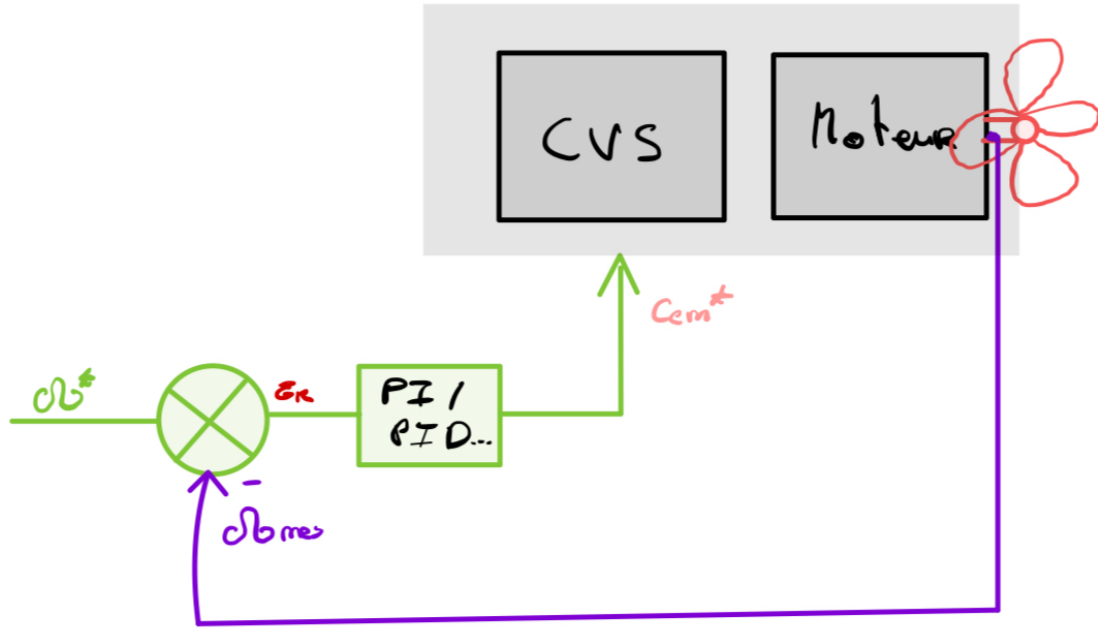


Figure 5.8: Global speed-control architecture implemented in the simulation environment.

For phase i ,

$$\varphi_i = \varphi_m \sqrt{2} \cos\left(\theta - (i-1)\frac{2\pi}{3}\right),$$

$$e_i = -\frac{d\varphi_i}{dt} = \frac{d\varphi_i}{d\theta} \frac{d\theta}{dt} = \sqrt{2}\varphi_m \omega \sin\left(\theta - (i-1)\frac{2\pi}{3}\right).$$

The electrical power relations are

$$P = 3VI \cos \alpha,$$

$$P_e = 3EI \cos \psi,$$

and

$$C_{em} = \frac{P_e}{\Omega} = 3 \frac{E}{\Omega} I \cos \psi.$$

If the current-to-EMF angle is maintained constant, the torque is proportional to the sinusoidal current amplitude. Adjusting this amplitude therefore provides direct torque control while preserving synchronism with the machine EMFs.

5.4.2 Ideal source and intersective PWM comparison

The SIMBA model VSI-vs-Perfect-V compares two supply configurations:

1. an ideal three-phase sinusoidal voltage source;
2. a voltage-source inverter controlled by intersective PWM.

The PWM converter generates a pulse train whose duty ratio varies with the sinusoidal reference. The ideal source contains only the fundamental component, whereas the PWM voltage includes switching harmonics concentrated around the carrier frequency and its multiples.

A second converter analysis confirms the same behaviour.

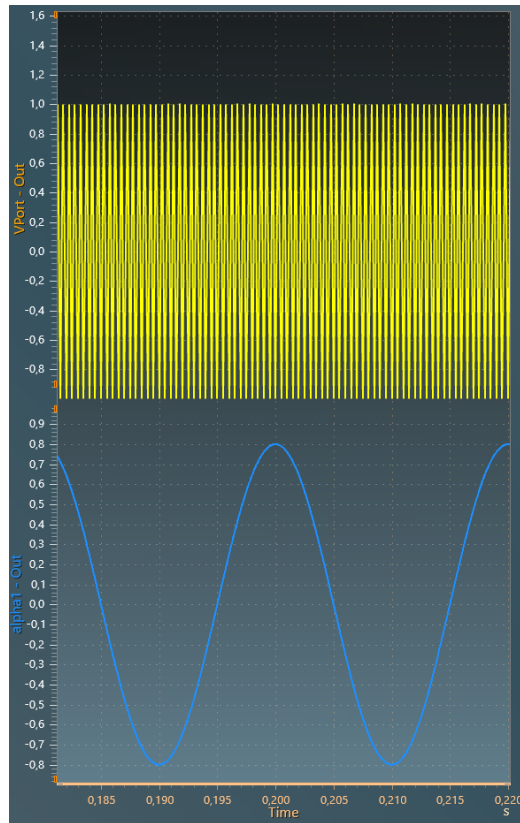


Figure 5.9: Representative ideal-source and PWM voltage waveforms.

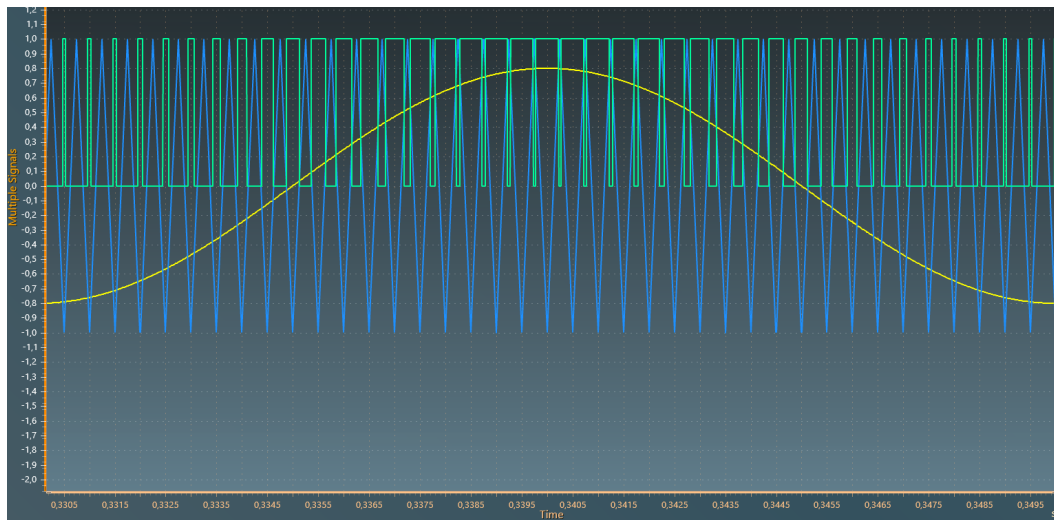


Figure 5.10: Time-domain and spectral comparison of the ideal and PWM voltages.

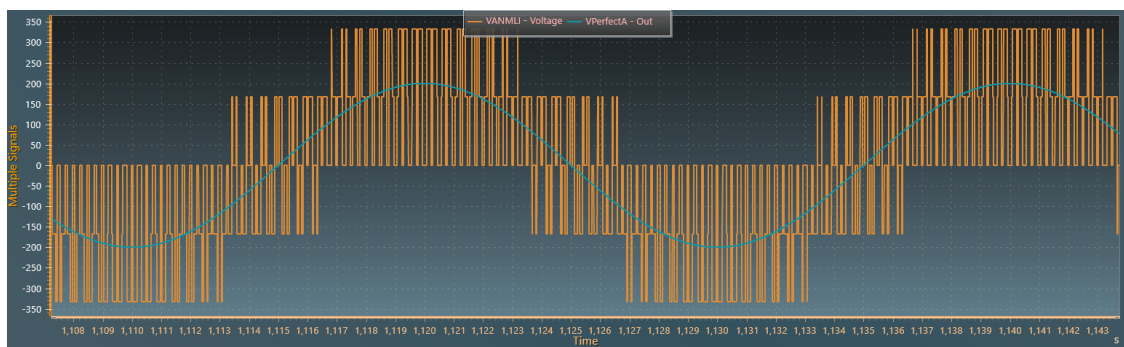


Figure 5.11: PWM phase voltage compared with the ideal sinusoidal phase voltage.

The fundamental remains at 50 Hz, while additional components appear around 2 kHz and 4 kHz for the selected carrier frequency.

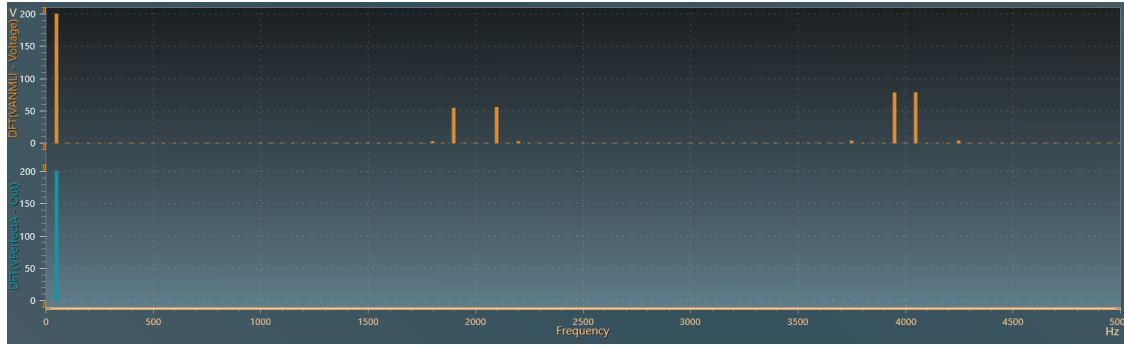


Figure 5.12: Discrete Fourier transform of the converter output.

The machine inductance attenuates the high-frequency voltage harmonics, but residual current ripple increases copper losses and can contribute to acoustic noise, torque ripple, and additional converter stress. Carrier frequency and filtering must therefore be selected jointly.

For the equivalent R-L phase circuit,

$$f_c = \frac{R}{2\pi L} = \frac{10}{2\pi \cdot 0.001} = 150 \text{ Hz},$$

$$\tau = \frac{L}{R} = \frac{0.001}{10} = 1 \text{ ms}.$$

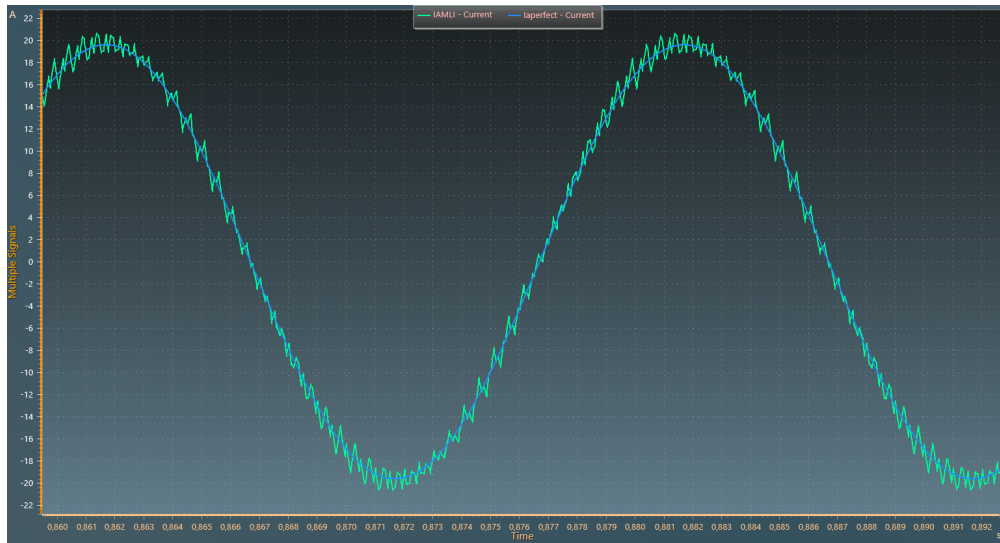


Figure 5.13: Ideal and PWM-fed phase-current comparison.

The ideal current is purely sinusoidal. The PWM-fed current contains ripple superimposed on the same fundamental. A DFT would therefore show an identical 50 Hz fundamental under correct scaling, with additional components clustered around multiples of the switching frequency.

5.4.3 Carrier-frequency and inductance study

The simulations were repeated with a carrier frequency of $f_{PWM} = 500 \text{ Hz}$ and $L_d = L_q = 0.01 \text{ H}$. The modified inductance reduces ripple, although a lower carrier frequency generally increases low-order current distortion and may reduce the accuracy of vector control.

The carrier-frequency trade-off is as follows:

- **high carrier frequency (2000 Hz):** lower current ripple and improved waveform fidelity, but higher semiconductor switching losses;

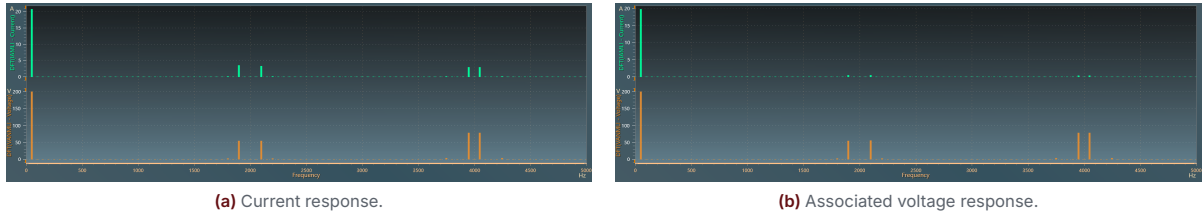


Figure 5.14: Modified carrier-frequency and inductance configuration.

- **low carrier frequency (500 Hz):** lower switching losses, but greater current ripple and stronger reliance on inductive filtering.

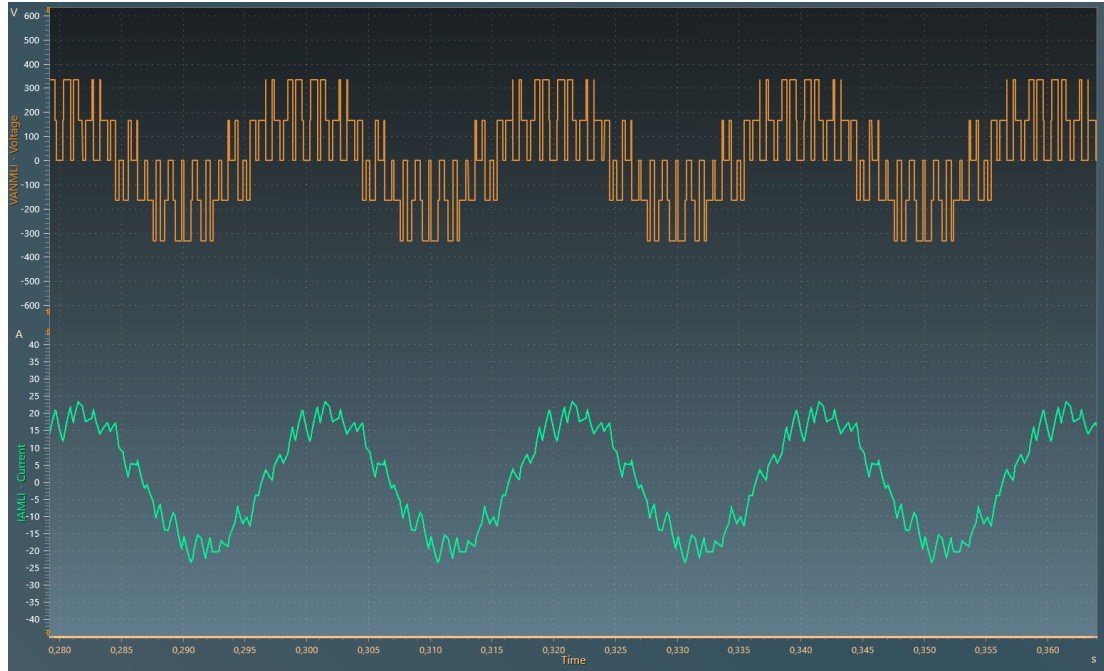


Figure 5.15: Comparison used to assess the influence of carrier frequency.

Returning to the initial values improves current quality because of the higher modulation frequency, but increases switching losses and thermal loading. Increasing V_{max} to 350 V expands the available voltage amplitude and converter dynamic range, while imposing higher electrical stress on the semiconductor devices and filters.

5.4.4 Closed-loop current control

Accurate phase-current control requires the applied voltage $V_k(t)$ to be adjusted in real time to compensate the resistive, inductive, and back-EMF terms while tracking $I_{k,ref}$. In the vector-control architecture:

1. a Park transformation maps I_a, I_b, I_c into the rotating $d-q$ frame;
2. PI/PID loops regulate i_d and i_q and generate V_d and V_q ;
3. PWM converts the voltage commands into switching signals for the inverter.

The $d-q$ transformation decouples the flux-producing current from the torque-producing current. This is essential for achieving fast torque control with limited losses.

The large matrix block performs the reference-frame transformation and produces the modulated output signal supplied to the converter.

During each acceleration phase, the electromagnetic torque reaches its maximum before settling once speed regulation is established. During deceleration, the torque becomes negative to provide controlled braking. The complete chain therefore demonstrates the expected relationship between speed error, torque demand, and the regulated $d-q$ currents.

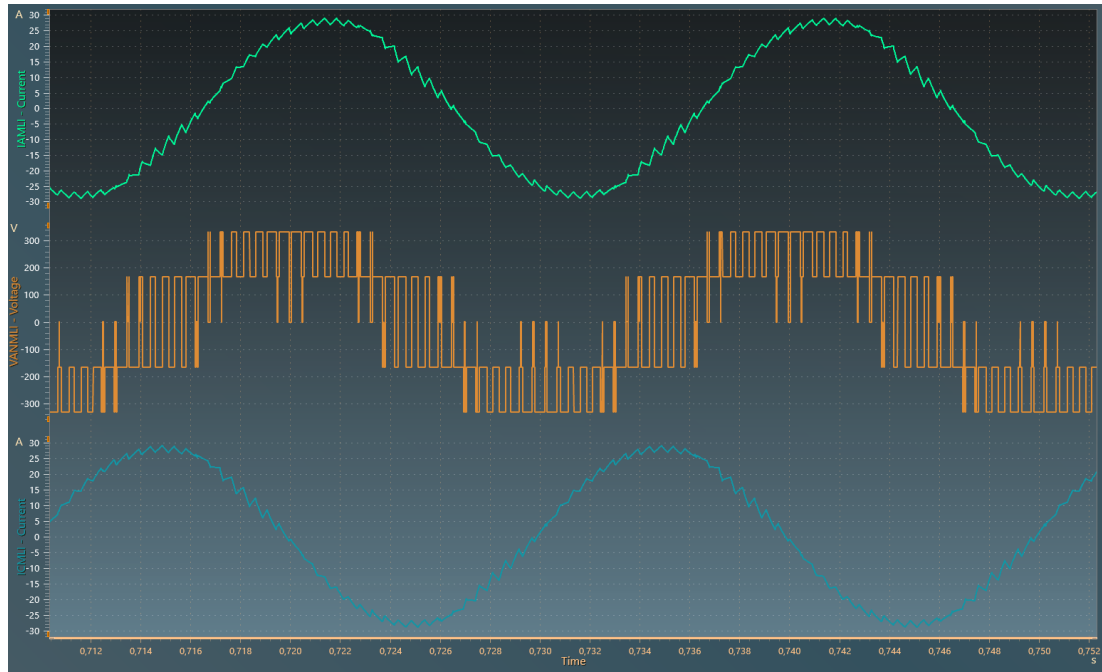
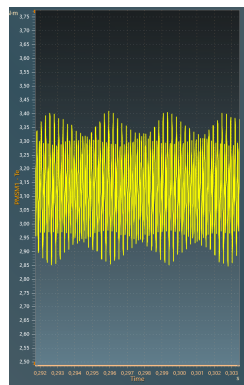
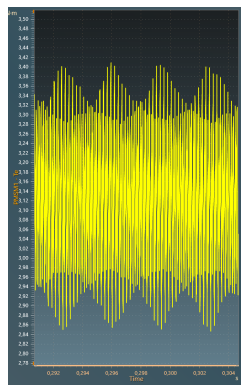


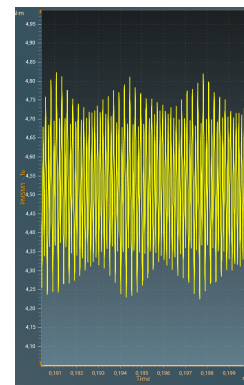
Figure 5.16: Voltage and current response after increasing the DC-link voltage limit to 350 V.



(a) Electromagnetic-torque block.

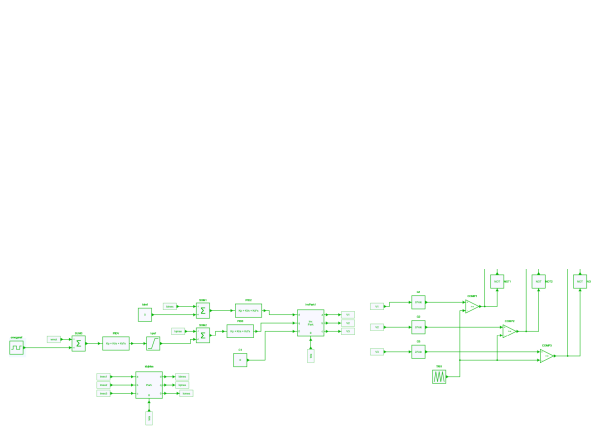


(b) Torque response for $\Psi = 0^\circ$.



(c) Responses for $\Psi = 45^\circ$ and 90° .

Figure 5.17: Torque-model implementation and sensitivity to the current-to-EMF angle.



(a) Matrix modulation and reference-frame transformations.



(b) Detailed current-control implementation.

Figure 5.18: Principal blocks of the closed-loop current-control model.

	Name	X1	Y1	X2	Y2	ΔX	1/ ΔX	ΔY	Max	Min	Peak to Peak	Average	RMS
	IC - Current	0.01	-18.727293	0.015	-18.290999	0.005	200	0.43629349	-17.975859	-19.079104	1.103245	-18.542199	18.538584
	IdqMes - D	0.01	0.036003068	0.015	0.032414485	0.005	200	-0.0035885831	0.81911564	-0.81362434	1.63274	0.0012559721	0.39884475
	IdqMes - Q	0.01	21.971778	0.015	21.97291	0.005	200	0.001131525	22.370195	21.582186	0.78800879	22.013141	22.00742
	PMSM1 - Te	0.01	9.8873002	0.015	9.8878094	0.005	200	0.00050918624	10.066588	9.7119837	0.35460395	9.9059136	9.903339
	omega - Out	0.01	2.9396417	0.015	4.4167791	0.005	200	1.4771374	4.4199055	2.9409848	1.4789207	3.6850516	3.7075416

Figure 5.19: Rotational-speed reference, measured speed, and electromagnetic torque.

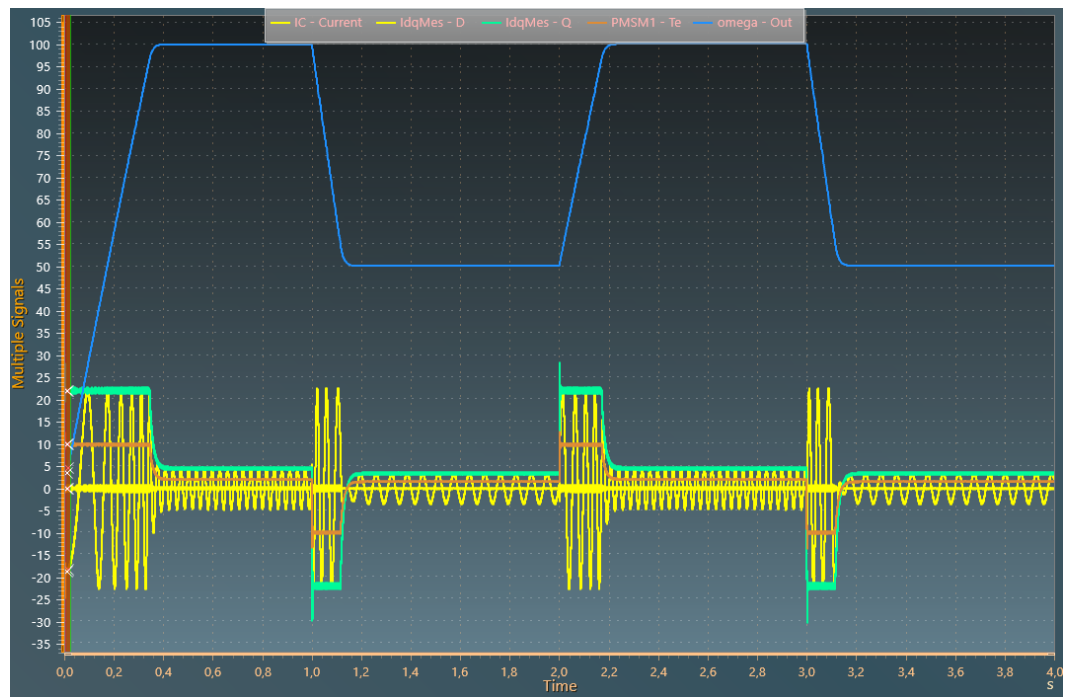


Figure 5.20: Current components and their references during the speed-control sequence.

Additive Manufacturing and Experimental Validation

6.1 Prototype preparation and manufacture

The final CAD model was imported into PrusaSlicer, oriented, supported, and divided into printable layers before manufacture by fused-filament fabrication.

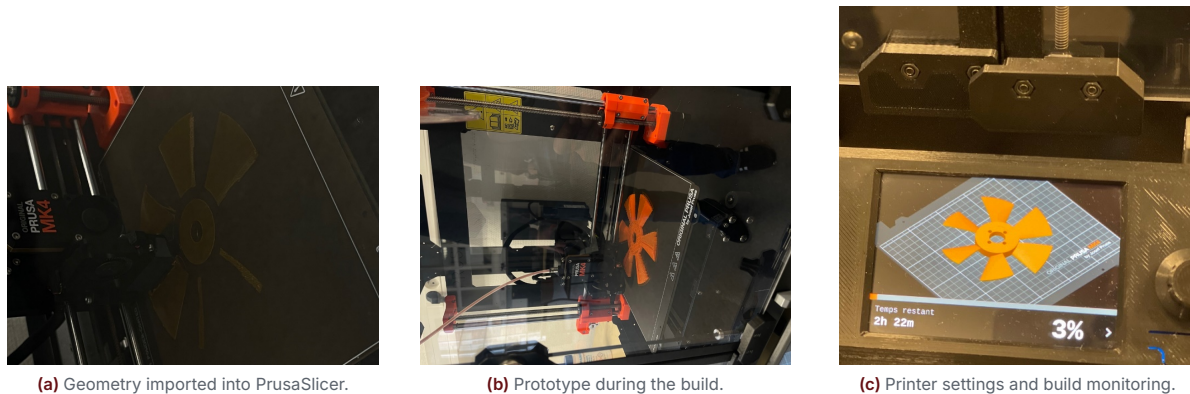


Figure 6.1: Preparation and execution of the fused-filament fabrication process.

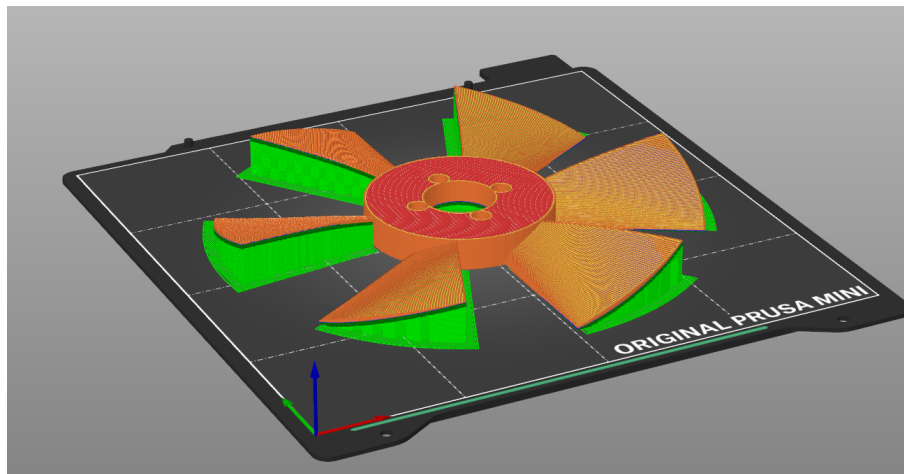


Figure 6.2: Sliced geometry and support strategy.

After the required supports had been removed, the complete prototype was obtained. The part was manufactured in polylactic acid (PLA), a widely used thermoplastic for rapid prototyping. The extrusion temperature was set to approximately 200°C. The slicer defined the layer sequence, toolpaths, support geometry, and local build parameters required to reproduce the complex blade and hub geometry.

The original manufacturing note reports a filament diameter of 20 μm . In practice, this value is more representative of a layer-scale or positioning parameter than of commercial filament stock; the reported setting is nevertheless retained here as part of the experimental record.

Type d'élément	Durée	Pourcentage	Filament utilisé
Périmètre	12m	6,5%	1.57 m 4.68 g
Périmètre externe	11m	6,0%	1.23 m 3.68 g
Périmètre en surplomb	14m	7,8%	0.43 m 1.29 g
Remplissage interne	7m	3,7%	1.16 m 3.45 g
Remplissage solide	27m	15,0%	4.73 m 14.11 g
Remplissage solide supérieur	2m	1,0%	0.19 m 0.57 g
Remplissage du pont	3m	1,6%	0.35 m 1.03 g
Supports	35m	19,0%	3.27 m 9.74 g
Interface des supports	41m	22,6%	2.43 m 7.24 g
Personnalisé	10s	0,1%	0.02 m 0.05 g

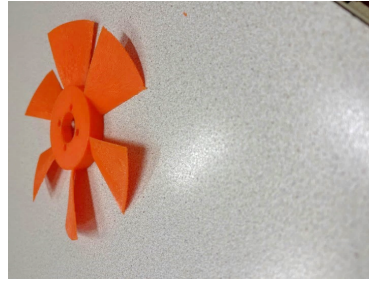
Temps d'impression estimés [Mode normal]:

Première couche: 14m

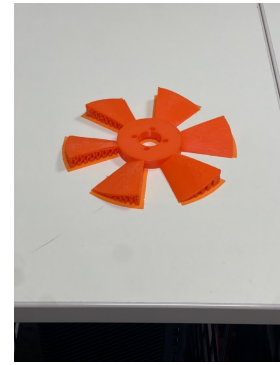
Total: 3h1m

Afficher le mode furtif

(a) Turbine before complete support removal.



(b) Runner and support-structure detail.

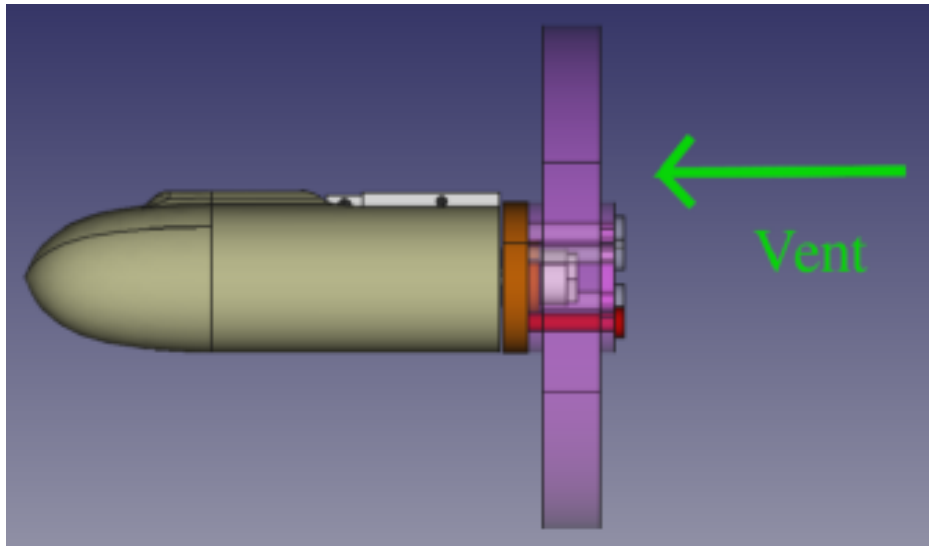


(c) Final prototype after support removal.

Figure 6.3: Post-processing stages of the additively manufactured turbine.

6.2 Wind-tunnel test facility

The prototype was tested in a recirculating wind tunnel as an aerodynamic analogue of the hydraulic machine. The purpose was not to reproduce water-turbine Reynolds numbers exactly, but to identify the qualitative performance curve, optimal rotational speed, and mechanical power trend under controlled flow conditions.

**Figure 6.4:** Installed turbine prototype in the wind-tunnel test section.

The facility is a return-circuit tunnel with a semi-guided test section and a maximum flow velocity of approximately 40 m/s. The contraction accelerates the flow by a factor of about 12.5, producing a relatively uniform low-turbulence stream in the measurement section. The return-valve region, by contrast, exhibits a turbulence level of approximately 10%.

6.3 Measured quantities and performance coefficients

The quantities used in the test analysis are summarised below.

Quantity	Definition	Expression or source
C_p	power coefficient	$P/(0.5\rho SV^3)$
P	mechanical power recovered at the shaft	$C\omega$
C	rotor torque	experiment
ω	rotational speed	$\pi N/30$
V	upstream reference velocity	experiment

Quantity	Definition	Expression or source
U	blade-tip circumferential speed	$R\omega$
λ	tip-speed ratio	U/V
S	rotor-disc area	$\pi(2R)^2/4$
R	blade-tip radius	75 mm
ρ	air density	1.2 kg/m ³
C_c	torque coefficient	derived from C_p

A first measurement series was performed at a wind speed of 10 m/s to identify the rotational speed corresponding to the maximum power coefficient. The optimum was found near $\omega = 190.5$ rad/s, with

$$C_{p,max} = 0.239.$$

A second series varied the upstream velocity to determine mechanical power as a function of flow speed. The wind velocity was obtained from the measured pressure difference using

$$V_{wind} = \sqrt{\frac{2\Delta P}{\rho_{air}}}.$$

The available aerodynamic power is

$$P_{available} = \frac{1}{2}\rho_{air}SV_{wind}^3.$$

The experimental power trend increases approximately linearly over the measured range and remains consistent with the expected qualitative behaviour. The maximum experimental efficiency is approximately 0.56. This is below the 0.80-0.90 range typically associated with mature industrial hydraulic turbines, but the comparison must be interpreted cautiously because the prototype was tested in air, at reduced scale, with additive-manufacturing surface roughness and without complete hydraulic similitude.

The experiment nevertheless validates the mechanical integrity of the prototype and confirms that the designed rotor develops a measurable torque and power maximum. It also identifies clear routes for improvement: reduced surface roughness, a more precise hub and tip clearance, higher-fidelity instrumentation, Reynolds-number scaling, and ultimately testing in a water loop.

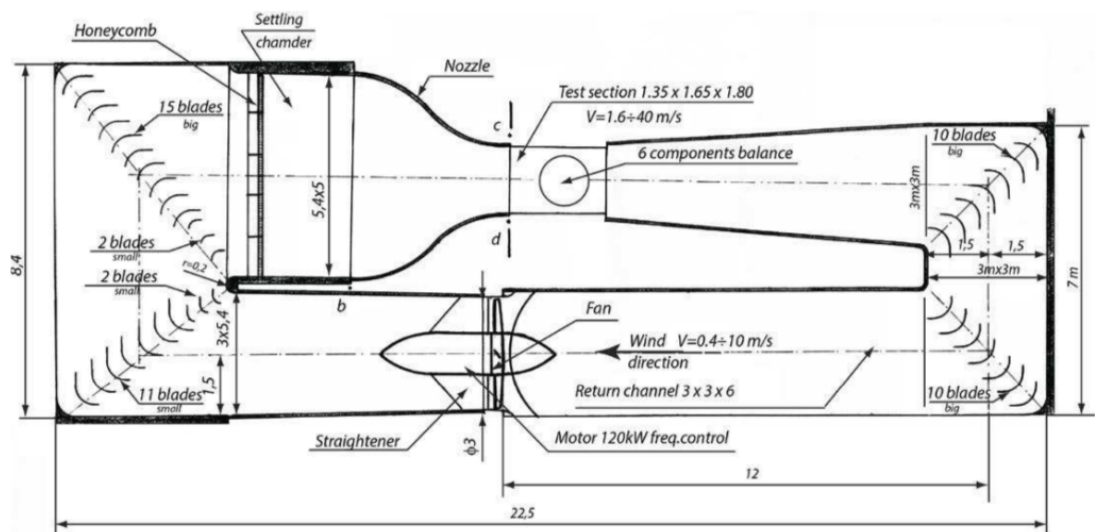
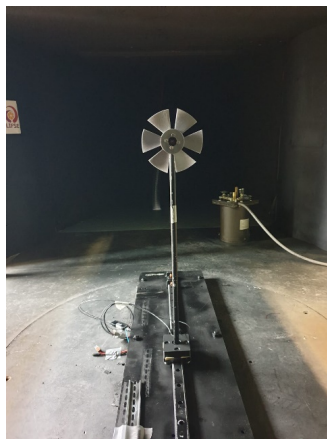
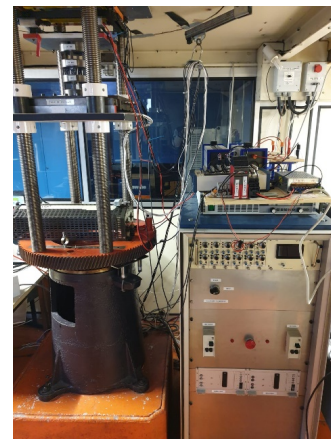


Figure 6.5: Schematic representation of the recirculating wind-tunnel facility.



(a) Prototype and instrumentation in the test section.



(b) Alternative view during the experiment.

Figure 6.6: Wind-tunnel installation of the turbine prototype.

Industrialisation and Market Deployment

7.1 Industrialisation framework

The industrial study translates the engineering concept into a production and deployment scenario. Hydro-electric power is treated as a mature renewable-energy option capable of supporting regional decarbonisation, reducing dependence on fossil generation, and providing long-lived infrastructure. The proposed industrialisation plan addresses not only technical performance, but also cost, capacity, logistics, sourcing, and market access.

The scope includes:

- market analysis and regional site selection;
- a sales and operations plan, referred to in the original study as the *Plan Industriel et Commercial* (PIC);
- a master production schedule (MPS/PDP);
- Manufacturing Resource Planning II (MRPII);
- a manufacturing bill of materials (MBOM);
- assembly, takt-time, lead-time, and plant-layout considerations;
- make-or-buy and country-location decisions;
- investment, profitability, and payback analysis.

7.2 Market analysis and regional selection

Several French regions were compared in terms of hydroelectric potential, energy demand, logistics, and infrastructure access.

The Provence-Alpes-Cote d'Azur (PACA) region was selected in the original business scenario for five reasons:

1. **Seasonal energy demand.** The regional population and electricity demand increase strongly during the summer period, making additional local generation strategically valuable.
2. **Hydroelectric potential.** Although lower than that of Auvergne-Rhone-Alpes, the available resource is considered sufficient for distributed projects while retaining logistical and operational advantages.
3. **Existing modernisation programmes.** Ongoing infrastructure renewal and renewable-energy initiatives create a favourable project environment.
4. **Logistical accessibility.** Ports, road networks, and industrial infrastructure reduce equipment-distribution and installation complexity.
5. **Environmental and economic co-benefits.** Local manufacture and deployment can support employment and contribute to carbon-reduction objectives.

The industrial objective is to become a supplier of modular hydroelectric equipment for institutional customers and infrastructure operators. The study assumes annual regional investment exceeding EUR 50 million and seeks to combine high product quality with controlled cost, seasonal flexibility, and scalable production.

7.3 Turbine selection and product positioning

A comparative assessment of turbine technologies was performed.

Critères	Auvergne-Rhône-Alpes	Occitanie	Grand Est	Nouvelle-Aquitaine	PACA
Potentiel hydraulique	9/10 (65 TWh/an)	7/10 (45 TWh/an)	6/10 (35 TWh/an)	8/10 (50 TWh/an)	8/10 (45 TWh/an, +20 % prévu)
Densité industrielle	8/10 (20 000 entreprises industrielles)	6/10 (12 000)	7/10 (15 000)	5/10 (10 000)	7/10 (15 000)
Réseaux logistiques	9/10 (2 ports, 1 aéroport, 5 lignes TGV)	7/10 (1 port, 1 aéroport, 2 lignes TGV)	8/10 (2 ports, 1 aéroport, 4 lignes TGV)	7/10 (1 port, 1 ligne TGV)	9/10 (3 ports, 2 aéroports, 2 lignes TGV)
Main-d'œuvre qualifiée	8/10 (Université Grenoble-Alpes, INSA Lyon)	6/10 (Université de Toulouse)	7/10 (Université de Strasbourg)	6/10 (Université de Bordeaux)	7/10 (Université Aix-Marseille)
Subventions potentielles	7/10 (5% réduction fiscale)	9/10 (8%)	6/10 (4%)	8/10 (6%)	8/10 (6%, soutien fort aux énergies renouvelables)
Impact environnemental	6/10 (Zones sensibles, contraintes élevées)	7/10 (Moins de pression)	5/10 (Zones protégées)	6/10 (Zones touristiques)	7/10 (Zones sensibles, mais projets adaptés)
Coût du terrain	7/10 (100 €/m ² en moyenne)	8/10 (80 €/m ²)	6/10 (120 €/m ²)	7/10 (90 €/m ²)	6/10 (150 €/m ²)
Score total	54/70	50/70	45/70	47/70	52/70

Figure 7.1: Comparison of potential French regions for turbine industrialisation and deployment.

The Kaplan principle was retained because the target sites combine low heads with relatively high and variable flow rates. The principal arguments are:

- suitability for heads below approximately 30 m and high discharge;
- high industrial efficiency, typically stated in the study as 90-95% under appropriate operating conditions;
- compatibility with modern regulation systems;
- robust operation over variable seasonal flow;
- potential for reduced environmental impact through controlled flow and blade-pitch operation;
- consistency with existing hydroelectric projects on rivers such as the Durance.

The study then compares large Kaplan installations with smaller modular units.

A compact bulb-turbine architecture is selected as the reference industrial product. This concept is well suited to low-head waterways, variable discharge, and modular deployment.

7.4 Sales and operations plan

The regional scenario targets an additional annual generation of approximately 2 TWh by 2030 through progressive installation of compact bulb turbines.

The reference unit is defined as follows.

Characteristic	Assumed value
nominal head	1.5-10 m

Characteristic	Assumed value
unit power	2 MW
maximum efficiency	85-90%
estimated cost per turbine	EUR 2 million
design life	30-40 years

The assumed yearly output per turbine is

$$2 \text{ MW} \times 6000 \text{ h} = 12 \text{ GWh/year.}$$

The number of turbines required for 2 TWh/year is therefore

$$\text{Number of turbines} = \frac{2000}{12} \approx 167.$$

The proposed five-year installation trajectory is:

Year	Turbines produced	Cumulative installed turbines	Cumulative annual production (GWh)
2025	33	33	396
2026	33	66	792
2027	33	99	1,188
2028	34	133	1,596
2029	34	167	2,004

The manufacturing concept includes five parallel assembly lines, each capable of completing one turbine approximately every two weeks. Critical runners and blades are machined locally, while certified suppliers provide generators and selected electronic systems.

The baseline staffing assumption is eight skilled operators and one supervisor per assembly line, supported by three maintenance technicians per site. The total annual production and maintenance workforce is approximately 60 people.

The principal raw-material assumptions per turbine are:

- 20 t of stainless steel;
- one generator and three principal sensors;
- biodegradable lubricants and associated auxiliary systems.

A safety stock equal to 10% of the cumulative installed fleet is proposed. This yields approximately three units in 2025 and 17 units in 2029.

7.5 Master production schedule

The master production schedule distributes 167 turbines over 60 months, with an average output of approximately three turbines per month.

The required production functions are:

- component machining;
- mechanical assembly;
- electrical-system integration;
- painting and surface treatment;
- quality control and acceptance testing.

The assumed installed resources include five CNC lines, three dedicated assembly stations, and two paint

Type de Turbine	Conditions d'utilisation	Efficacité	Coût (€/kW installé)	Adaptabilité aux projets PACA	Avantages principaux	Inconvénients
Kaplan	Faibles hauteurs de chute (<30 m) et débits élevés	90-95 %	1 200 - 2 000	Excellente	Haut rendement avec des débits variables	Plus complexe à installer et à maintenir
Pelton	Grandes hauteurs de chute (>100 m) et débits faibles	85-90 %	1 500 - 2 500	Faible	Idéale pour les régions montagneuses	Non adaptée aux faibles hauteurs de chute
Francis	Hauteurs de chute modérées (30-300 m), débits variés	80-90 %	1 000 - 1 800	Modérée	Bonne polyvalence pour des conditions variées	Moins efficace pour les très faibles hauteurs
Bulbe	Très faibles hauteurs de chute (<15 m) et débits élevés	80-85 %	1 800 - 2 500	Modérée	Idéale pour les zones avec très faibles pentes	Rendement inférieur à celui des Kaplan
Vis d'Archimède	Très faibles hauteurs de chute et débits très faibles	70-80 %	2 500 - 3 000	Faible	Simplicité de conception	Rendement faible, adapté uniquement aux microcentrales

Figure 7.2: Comparison of turbine technologies for the selected regional conditions.

Critère	Turbine Kaplan (grande capacité)	Petites turbines (Kaplan compactes, Bulbe, etc.)
Coût initial	Élevé (1 200 - 2 500 €/kW)	Faible à modéré (500 - 1 500 €/kW)
Maintenance	Optimisée avec technologies modernes	Maintenance accrue avec plusieurs unités
Rendement	90-95 %	75-85 %
Modularité	Faible	Excellente
Durée de vie	Longue (40-50 ans)	Modérée (20-30 ans)
Impact environnemental	Moyen	Faible
Adaptation aux débits	Meilleure pour grands débits	Meilleure pour faibles débits

Figure 7.3: Large Kaplan units compared with smaller modular turbines.

Caractéristique	Valeur ou Plage Typique
Hauteur de chute nominale	1,5 m à 10 m
Débit optimal	5 m ³ /s à 100 m ³ /s
Puissance installée	100 kW à 3 MW (par unité)
Rendement maximal	85-90 %
Type de régulation	Régulation simple, avec palettes fixes ou ajustables
Durée de vie	30-40 ans (avec maintenance régulière)
Dimensions compactes	Installation intégrée dans des conduits ou canaux existants
Matériaux	Acier inoxydable et alliages résistants à la corrosion
Technologie additionnelle	Surveillance numérique pour le contrôle à distance et maintenance prédictive

Figure 7.4: Compact bulb-turbine concept adopted for the industrial scenario.

Données	2025	2026	2027	2028	2029
Prévisions de vente (unités)	33	66	99	133	167
Nombre d'unités produites (unités)	33	66	99	133	167
Stock final (unités)	33	33	33	34	34
Stock de sécurité (unités)	3	7	10	13	17

Figure 7.5: Sales and operations plan, including annual production and safety stock.

booths. Planning is organised at three horizons: weekly scheduling for the first 12 months, quarterly review from 12 to 36 months, and annual adjustment from 36 to 60 months.

Year	Annual quantity	Quarterly quantity	Average monthly quantity
2025	33	8-9	2.75
2026	33	8-9	2.75
2027	33	8-9	2.75
2028	34	8-9	2.83
2029	34	8-9	2.83

For stainless steel, the total requirement is

$$20 \text{ t/turbine} \times 167 = 3340 \text{ t.}$$

The original plan assumes quarterly steel deliveries of approximately 167 t, monthly electronic-system deliveries, and 50 kg of paint and surface-treatment products per turbine.

The labour model is summarised below.

Operation	Required operators	Hours per turbine
component machining	20	50
assembly	30	40
painting and surface treatment	10	20
quality control	10	10
Total stated in the original schedule	85	120

The operator total shown in the source schedule is retained, although the listed row values sum to 70. This

inconsistency should be resolved before an investment decision.

Key performance indicators include monthly output versus plan, defect rate with a target below 2%, and supplier delivery performance. Monthly production reviews and quarterly programme reviews are proposed.

7.6 MRPII, MBOM, assembly, and plant flow

The MRPII assumptions are:

- annual cadence: 33-34 turbines;
- working time: 35 hours per operator per week;
- two parallel teams;
- 220 working days per year;
- average stated manufacturing time: 150 hours per turbine.

For an average of three turbines per month, the required monthly operation times are:

- machining: $50 \times 3 = 150$ h/month;
- assembly: $40 \times 3 = 120$ h/month;
- painting and treatment: $20 \times 3 = 60$ h/month;
- quality control: $10 \times 3 = 30$ h/month.

One operator provides approximately

$$35 \text{ h/week} \times 4 = 140 \text{ h/month.}$$

The manufacturing bill of materials contains four principal families:

1. **Main structure:** turbine runner, main shaft, and casing.
2. **Moving components:** fixed and moving blades and rotational bearings.
3. **Electrical systems:** generator, control system, and automation hardware.
4. **Auxiliaries:** lubrication, anchors, and fasteners.

The reference per-turbine quantities are 20 tonnes of stainless steel, ten blades, 20 litres of biodegradable lubricant, one generator, and three sensors.

Assembly is divided into pre-assembly, final assembly, and testing. The runner and blades are assembled first, followed by the casing. The main shaft, bearings, generator, and auxiliary systems are then integrated. Mechanical checks, electrical calibration, and performance testing complete the process.

The available monthly production time for two teams is

$$35 \text{ h/week} \times 4 \text{ weeks/month} \times 2 = 280 \text{ h/month.}$$

For a demand of three turbines per month, the takt time is

$$\text{Takt time} = \frac{280}{3} = 93.33 \text{ h/turbine.}$$

The stated direct lead time is

$$50 + 40 + 20 + 10 = 120 \text{ h/turbine.}$$

Because the lead time exceeds the takt time, the plan relies on parallel resources and overlapping operations. This relationship should be tested with a discrete-event model before finalising the line capacity.

The intended production flow is:

1. receipt of steel and electrical systems;
2. machining of metallic components;
3. pre-assembly of subassemblies;
4. transfer to final assembly;

5. painting and surface treatment;
6. final test;
7. central storage and delivery to the installation site.

The cost breakdown retained in the study is:

Cost item	Cost per turbine	Share
machining	EUR 0.8 million	40%
assembly	EUR 0.6 million	30%
electrical systems	EUR 0.4 million	20%
surface treatment and quality	EUR 0.2 million	10%
Total	EUR 2.0 million	100%

For 167 turbines,

$$167 \times \text{EUR } 2 \text{ million} = \text{EUR } 334 \text{ million.}$$

The plant layout places CNC machines close to raw-material storage, aligns assembly stations to minimise transfers, isolates paint booths for environmental compliance, and centralises critical-component and finished-product storage.

7.7 Make-or-buy strategy

The make-or-buy analysis distinguishes performance-critical parts from standard or specialist components.

Component	Make	Buy	Rationale
turbine runner	internal machining	-	direct control of quality and geometry, at higher capital cost
fixed and moving blades	internal production	-	dimensional flexibility and performance control
casing	-	local supplier	lower cost, with supplier dependency
electrical generator	-	qualified supplier	access to specialist capability, with lead-time risk
control systems	-	specialist partners	expertise and certification requirements
lubrication system	internal integration	-	environmental and system-level control

The recommended strategy is mixed: manufacture the runner, blades, and other performance-critical elements internally; source standard or specialist components such as generators, casings, and control electronics.

7.8 Country-location strategy

The location criteria are labour cost, access to raw materials, industrial infrastructure, proximity to target sites, and political and economic stability.

Country	Advantages	Disadvantages
France	proximity to installation sites; strong industrial expertise and quality control	high labour cost
Poland	competitive labour cost; developed industrial infrastructure	moderately more complex logistics
Turkey	highly competitive metal-processing cost; good access to steel	greater transport distance
China	very low cost for standard components; extensive industrial capacity	long lead times, quality risk, and trade exposure
Portugal	balanced labour cost and European proximity	somewhat lower heavy-industrial capacity

The recommended configuration retains critical machining in France and outsources selected standard components to Poland or Turkey. The source study estimates approximately EUR 2.5 million per turbine for an all-France solution and EUR 2.0 million for the mixed strategy.

7.9 Profitability analysis

The commercial assumptions are:

- selling price: EUR 3 million per turbine;
- total sales: 167 turbines;
- total revenue: $167 \times 3 = \text{EUR } 501 \text{ million}$;
- manufacturing cost: EUR 2 million per turbine;
- total manufacturing cost: $167 \times 2 = \text{EUR } 334 \text{ million}$;
- R&D and initial investment: EUR 30 million;
- fixed costs over five years: EUR 50 million.

The gross margin per turbine is

$$\text{EUR } 3 \text{ million} - \text{EUR } 2 \text{ million} = \text{EUR } 1 \text{ million.}$$

The total gross margin is

$$167 \times \text{EUR } 1 \text{ million} = \text{EUR } 167 \text{ million.}$$

The total net profit is

$$\text{EUR } 501 \text{ million} - (\text{EUR } 334 \text{ million} + \text{EUR } 30 \text{ million} + \text{EUR } 50 \text{ million}) = \text{EUR } 87 \text{ million.}$$

The return on investment is

$$ROI = \frac{\text{Net profit}}{\text{Total investment}} \times 100,$$

$$ROI = \frac{87}{334 + 30 + 50} \times 100 \approx 21.5\%.$$

With an annual gross margin of approximately

$$33 \text{ turbines/year} \times \text{EUR } 1 \text{ million/turbine} = \text{EUR } 33 \text{ million/year,}$$

the payback period on the EUR 80 million of R&D and fixed costs is

$$\text{Payback} = \frac{30 + 50}{33} \approx 2.4 \text{ years.}$$

Indicator	Value
total revenue	EUR 501 million
total cost	EUR 414 million
total net profit	EUR 87 million
ROI	21.5%
payback period	2.4 years

Under these assumptions, the project is profitable, with a gross margin of EUR 1 million per turbine and a relatively short payback period. The result remains highly sensitive to sales volume, manufacturing cost, installation scope, financing, warranty exposure, and the realism of the 6000 h/year utilisation assumption. A subsequent investment-grade study should therefore include a discounted cash-flow model, uncertainty ranges, and scenario analysis.

Conclusion

This project demonstrates a complete multidisciplinary workflow for the design, analysis, prototyping, and industrial positioning of a compact axial hydraulic turbine.

The hydraulic pre-design established a 1500 rpm operating point, a design head of 3.8 m, a volume flow rate of $0.132 \text{ m}^3 \cdot \text{s}^{-1}$, and a runner outer radius of 0.103 m. MATLAB and the Cordier diagram provided the principal similarity parameters, while the velocity-triangle calculations supplied the blade inlet and outlet angles. A modified NACA thickness law was fitted to obtain a blade profile with improved mechanical robustness.

The analytical geometry was transferred to CATIA V5, where the blade surfaces, runner, hub, duct, and fluid domains were constructed. A preliminary finite-element analysis confirmed that the largest stresses occur at the blade root and remain within the assumed elastic limit. STAR-CCM+ simulations then provided a first assessment of the velocity and pressure fields, convergence behaviour, power, efficiency, and cavitation-sensitive regions. The best simulated point delivered approximately 1760 W at $0.144 \text{ m}^3 \cdot \text{s}^{-1}$, indicating that the original 5 kW target would require either a redesigned runner, different operating conditions, or a multi-unit architecture.

The electromechanical study derived the induced phase voltages and electromagnetic torque of a surface-mounted permanent-magnet synchronous generator. Magnet thickness, fundamental air-gap flux density, current loading, thermal constraints, and active length were linked in a consistent first-order model. The associated voltage-source inverter study showed the expected compromise between PWM carrier frequency, current ripple, semiconductor switching losses, and filter requirements. Vector current control in the d - q frame provided the required decoupling between flux and torque production.

The additive-manufacturing stage converted the CAD model into a physical PLA prototype. Wind-tunnel testing identified a maximum power coefficient of approximately 0.239 and a maximum experimental efficiency of approximately 0.56. These values are not directly equivalent to full-scale hydraulic performance, but they validate the existence of a measurable optimum and highlight the importance of surface finish, scale effects, clearances, and dynamic similitude.

Finally, the industrial analysis translated the concept into a modular deployment scenario based on 167 compact bulb turbines and a target generation of roughly 2 TWh/year. The proposed production plan, make-or-buy strategy, and mixed France-European sourcing model yield a stated ROI of 21.5% and a payback period of 2.4 years under the source assumptions. The exercise reveals both the potential of the concept and the need for stricter reconciliation of production times, staffing totals, commercial scope, and financial uncertainty before industrial commitment.

The principal value of the work lies in the continuity of the engineering chain: similarity-based hydraulic sizing informs the blade geometry; CAD and structural analysis prepare the numerical flow model; CFD informs the generator and prototype stages; and the resulting technical definition supports production and business planning. Future work should prioritise a refined transient CFD model, water-loop testing, structural fatigue analysis, generator finite-element validation, and a scenario-based industrial business case.

MATLAB and XFoil Routines

The principal routines used for hydraulic preliminary sizing, blade mean-line construction, profile fitting, and point-cloud export are reproduced below. Comments have been translated into English; the mathematical operations and numerical coefficients are retained from the source implementation.

```

1  clear all; close all; clc
2
3  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
4
5  %Preliminary sizing of an axial turbine
6
7  %   - Complete the following script and the functions it calls
8
9  %   - Execute line by line using the F9 key
10
11 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
12
13 % options
14
15 Nom='marim';%Turbine name
16
17 %constants
18
19 g=9.81; %m^2/s
20
21 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
22
23 %Design specification
24
25 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
26
27 N=1500;%rpm
28
29 %Q1: Determine the angular velocity
30
31 omega=N*2*pi/60;%rad/s
32
33 Re=0.103;% penstock radius and turbine outer radius
34
35 Nsq=200;%engineering specific speed
36
37 %Q2: Determine the angular velocity
38
39 vas=Nsq*2*pi/60/g^.75;%specific angular velocity
40
41 disp(vas);
42
43 vas = Nsq*2*pi/60/g^.75; % specific speed
44
45 %% Sizing using the Cordier diagram
46
47 % q1 determine the specific radius from the Cordier diagram
48
49 Lambda = 0.701;
50
51 % q2 - Determine the head parameters (H and qv) as functions of
52 % vas, Lambda, and R

```

```

53 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
54 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
55
56 % graphical solution for H
57
58 H = 0.1:0.1:10; % definition of a variable
59
60 A = vas/omega*(g*H).^0.75 - Re/Lambda*(g*H).^0.25; % exploitation
61 % of the two-equation system
62
63 figure(2);
64
65 plot(H, A, 'b');
66
67 line([0 10], [0 0]);
68
69 xlabel('H (m)');
70
71 ylabel('A');
72
73 clear H;
74
75 %therefore
76
77 H=3.8;% design head
78
79 % deduce qv
80
81 qv=(Nsq*H^0.75/N)^2;
82
83 disp(qv);
84
85 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
86
87 %% Global preliminary sizing of the runner
88
89 Za = 6; % number of blades
90
91 eta = 0.92; % hydraulic efficiency
92
93 % aspect ratio T = Ri/Re:
94
95 T = 0.38;
96
97 Ri = Re*T;
98
99 Ri = floor(1000*Ri)/1000; % rounded to the nearest millimetre
100
101 disp('Re (m), Ri (m)');
102
103 disp([Re Ri]);
104
105 % q5: Create a series of five radii from Ri to Re to discretise
106
107 % the blade angle (rounded to the nearest millimetre)
108
109 nr = 5; % number of radial discretisation points
110
111 r = linspace(Ri, Re, nr);
112
113 r = floor(r*1000)/1000;
114
115 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
116
117 % Blade velocity triangles and runner-blade plotting
118

```

[illegible]

```

185
186 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
187
188 %Discretisation of the unit axial-extent mean line xx
189
190 %Refine near the leading edge
191
192 %arbitrary discretisation that may be modified
193
194 xba=[0,logspace(-6,-2,10)];% leading-edge region
195
196 xmc=linspace(0.011,0.1,9);% mid-chord region
197
198 xbf=linspace(0.11,1,9);% trailing-edge region
199
200 xx=[xba xmc xbf];
201
202 % Calculate the mean line using the ligneK function
203
204 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
205
206 %Q9: Edit and complete the LigneK script before executing the
207
208 %following command
209
210 [yy,beta_i]=ligneK(beta_1,beta_2,xx);
211
212 %Plot the unit axial-extent mean line
213
214 figure (2); hold on;
215
216 plot(xx,yy,'r');xlabel('xx[ ]');ylabel('yy[ ]');
217
218 legend('unit axial-extent mean line');
219
220 axis equal
221
222 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
223
224 %Thickness distribution along the mean line
225
226 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
227
228 %Q10: Set the radial variation of relative thickness tt(r) from the root
229 %of the blade (arbitrarily 13%) to the tip (arbitrarily 3%).
230 %
231
232 %This distribution may be improved at the end of the practical session
233
234 ep_pied=13/100;
235
236 ep_peri=3/100;
237
238 tt=linspace(ep_pied,ep_peri,nr);
239
240 figure (3);hold on;
241
242 legend('thickness law')
243
244 plot(r,tt);
245
246 xlabel('r(m)');ylabel('ep(%)');
247
248 %tt=[0.16 0.11 0.08 0.05 0.03];% an alternative tt(r) variation
249
250 %Q11: Answer Questions Q11.1 and Q11.2. Edit the epaisseurK function

```

```

251      %and complete its script to answer Question Q11.3
252
253      [xp1 yp1 xp2 yp2]=epaisseurK(xx,yy,beta_i,tt(u));
254
255      %transpose into column vectors
256
257      xp1=xp1'; yp1=yp1';
258
259      xp2=xp2'; yp2=yp2';
260
261      figure (4); hold on;
262
263      plot(xp1,yp1,'b')
264
265      plot(xp2,yp2,'g')
266
267      axis equal
268
269      %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
270
271      %Blade-row solidity
272
273      %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
274
275      %Q12: Recall the definition and role of solidity, including its benefits and drawbacks.
276
277      %drawbacks of high solidity.
278
279      serrage=0.6;
280
281      lx=serrage*(2*pi*r(u)/6);%chord length
282
283      xp1=xp1*lx; yp1=yp1*lx;%scale the profiles
284
285      xp2=xp2*lx; yp2=yp2*lx;
286
287      zz=ones(size(xx')).*r(u);%for 3D plotting
288
289      %for visualisation
290
291      figure (5); hold on;
292
293      plot3(xp1,yp1,zz,'r')
294
295      plot3(xp2,yp2,zz,'k')
296
297      %Additionally, plot the Ri and Re envelope circles
298
299      %angular discretisation
300
301      theta=0:0.01:2*pi;
302
303      yri=Ri*cos(theta);
304
305      zri=Ri*sin(theta);
306
307      xri=zeros(1,size(theta,2));
308
309      plot3(xri,yri,zri);
310
311      plot3(xri,yri/Ri*Re,zri/Ri*Re);
312
313      axis equal
314
315      view(120,16)
316

```



```

383
384     %Generate an export file for CATIA
385
386     Nom='Courbes-GSD';%Excel file name
387
388     TableauGSD={'StartLoft','',''};
389
390     for u=1:nr
391
392         %upper surface
393
394         TableauGSD=[TableauGSD ; {'StartCurve','',''}];
395
396         for k=1:length(XXX.ext(u).R)
397
398             TableauGSD=[TableauGSD ;
399 {XXX.ext(u).R(k,1),XXX.ext(u).R(k,2),XXX.ext(u).R(k,3)}];
400
401         end
402
403         TableauGSD=[TableauGSD ; {'EndCurve','',''}];
404
405         %lower surface
406
407         TableauGSD=[TableauGSD ; {'StartCurve','',''}];
408
409         for k=1:length(XXX.int(u).R)
410
411             TableauGSD=[TableauGSD ;
412 {XXX.int(u).R(k,1),XXX.int(u).R(k,2),XXX.int(u).R(k,3)}];
413
414         end
415
416         TableauGSD=[TableauGSD ; {'EndCurve','',''}];
417
418     end
419
420     TableauGSD=[TableauGSD ; {'EndLoft','',''}];
421
422     TableauGSD=[TableauGSD ; {'End','',''}];
423
424     xlswrite(Nom,TableauGSD); % copy these data into the CATIA file
425     % GSD_PointSplineLoftFromExcel.xls, which contains the import macro.

```

Listing A.1: Axial-turbine preliminary sizing and CATIA point-cloud export routine.

```

1  %Find an analytical expression for profiles whose
2
3  %trailing edge is thickened to ensure mechanical strength.
4
5  % Input: point file generated by XF0IL ("psave")
6
7  % Tool: MATLAB fminsearch or lsqcurvefit if the Optimization Toolbox is available.
8
9  clear all; close all; clc;
10
11 %Q11.21: enter the filename and corresponding relative thickness
12
13 fname='NACA0015_DEV_KUMAR.txt';
14
15 tt=0.15;%relative thickness of the profile to load
16
17 data=load(fname);%load the file
18
19 figure(1);hold on;%plot raw points

```

```

20
21 plot(data(:,1),data(:,2));
22
23 axis equal
24
25 %Q11.22: Define xdata and ydata for the upper surface (first half of the file points).
26
27 xdata=data(1:ceil(239/2),1);
28
29 ydata=data(1:ceil(239/2),2);
30
31 plot(xdata,ydata,'g-');% plot in red
32
33 %Q11.23: Plot the unmodified NACA thickness law; its coefficients initialise the fit.
34
35 naca0=[0.2969 -0.1260 -0.35116 +0.2843 -0.1015]; % standard NACA coefficients
36
37 % define the fonc_naca function with:
38
39 % tt=relative thickness, %
40
41 % x=chordwise abscissae
42
43 % Variables are defined inside @(), followed by the function expression.
44
45 fonc_naca=@(naca,x) tt/0.2*(naca(1)*x.^0.5+naca(2)*x+naca(3)*x.^2+naca(4)*x.^3+naca(5)*x.^4);
46
47 y0=fonc_naca(naca0,xdata); % recover the thickness distribution from naca0
48
49 plot(xdata,y0,'m-');%plot in green
50
51 %Q11.24: Configure fminsearch to obtain the new coefficients.
52
53 % The new ecartq function (sum of squared differences between
54 % data and model) is minimised to define the coefficients of the
55
56 % modified profile
57
58 ecartq= @(naca) sum((ydata-fonc_naca(naca,xdata)).^2);
59
60 naca_mod=fminsearch(ecartq, naca0);%See the fminsearch documentation
61
62 disp(naca_mod);%display the coefficients
63
64 plot(xdata,fonc_naca(naca_mod,xdata),'r*:') % zoom in to compare data and model

```

Listing A.2: Modified NACA thickness-law identification routine.

```

1 function [yy,beta_i]=ligneK(beta_1,beta_2,xx)
2
3 %plots the mean lines yy=f(xx) of the blade sections:
4
5 %with:
6
7     % inputs:
8
9     % beta_1:inlet angle
10
11     % beta_2:outlet angle
12
13     % xx normalised abscissae
14
15     % outputs:
16
17     % yy ordinates

```

```

18 % beta_i angles
19
20
21 %boundary conditions:
22
23 % linear variation of local angle beta_i between beta_1 and beta_2
24
25 %define:
26
27 % dyy local variation
28
29 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
30
31 %Q :Define dyy=f(xx), then yy=f(xx)
32
33 dyy=(tand(beta_2)-tand(beta_1)).*xx+tand(beta_1);
34
35 yy=0.5*(tand(beta_2)-tand(beta_1)).*xx.^2+tand(beta_1).*xx;
36
37 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
38
39 %Q :Define the local angle beta_i=f(dyy)
40
41 beta_i=atand(dyy);
42
43 end

```

Listing A.3: Blade-section mean-line generator.

[illegible]

```

37 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
38
39 % Calculate local chord length cc for scaling
40
41     cc=(xx(end)^2+yy(end)^2).^0.5;
42
43 % Calculate the coordinates of each envelope point
44
45     %upper surface xp1, yp1
46
47     xp1=xx-cc*yt.*sind(bei);
48
49     yp1=yy+cc*yt.*cosd(bei);
50
51     %lower surface: xp2, yp2
52
53     xp2=xx+cc*yt.*sind(bei);
54
55     yp2=yy-cc*yt.*cosd(bei);
56
57     % normalise by the chord length
58
59     xp1=xp1/cc;
60
61     yp1=yp1/cc;
62
63     xp2=xp2/cc;
64
65     yp2=yp2/cc;
66
67 end

```

Listing A.4: Blade-thickness distribution about the mean line.

A.1 Software environment and model scope

The study combines MATLAB for numerical sizing and data processing, XFOIL for section-profile manipulation, CATIA V5 for surface and solid modelling, STAR-CCM+ for computational fluid dynamics, SIMBA for the electromechanical drive and PWM investigations, and PrusaSlicer for additive-manufacturing preparation. Each numerical stage is a preliminary engineering model and should be interpreted within the assumptions stated in the corresponding chapter. In particular, the steady moving-reference-frame CFD model, the two-dimensional magnetic approximation, the wind-tunnel similarity experiment, and the high-level industrial business case are complementary design tools rather than substitutes for full-scale qualification.

Integrated Turbomachinery Design



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